

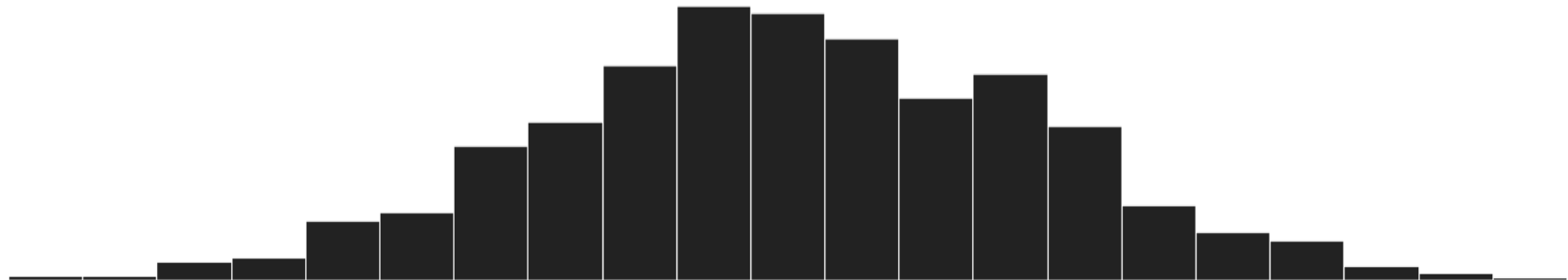
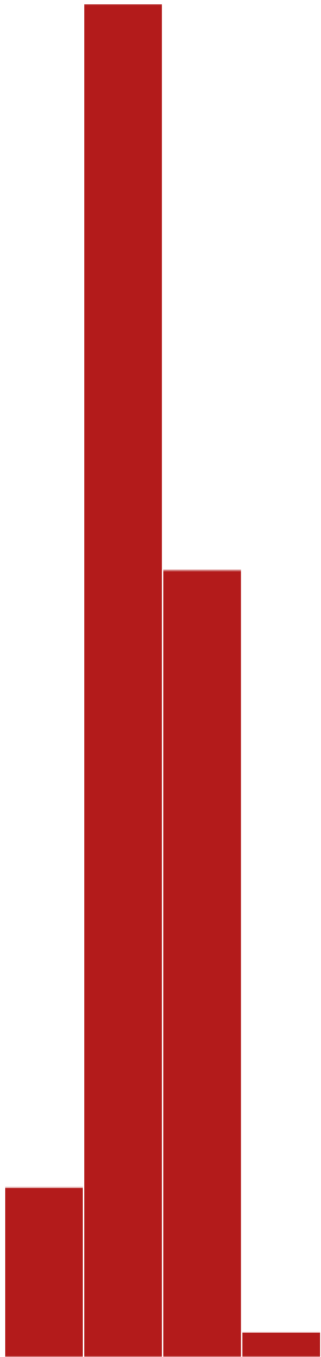
VTPEH 6105

Lecture 02

INFERENCE: PROBABILITIES & UNCERTAINTY DATA WRANGLING WITH

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House Keeping

- Office hours
 - **Tuesdays, 09:00-10:00 AM** (Jordan, Madison and Harsh)
 - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal and Daniel)
 - If you cannot make it at these times, please contact the course assistant of your choice by email
 - in S1-122, unless indicated otherwise on Canvas
- [VTPEH 6108 Topic Suggestion](#) – Any topic you would like to chat about? Drop a message on the dedicated discussion (anonymous). We'll do our best to accommodate.

Questions

- How to pick the best colors for your plots?



Version 1.0

Choose Your Colors

Select up to 4 colors for exploration and visualization.

Color 1:
#FF0000

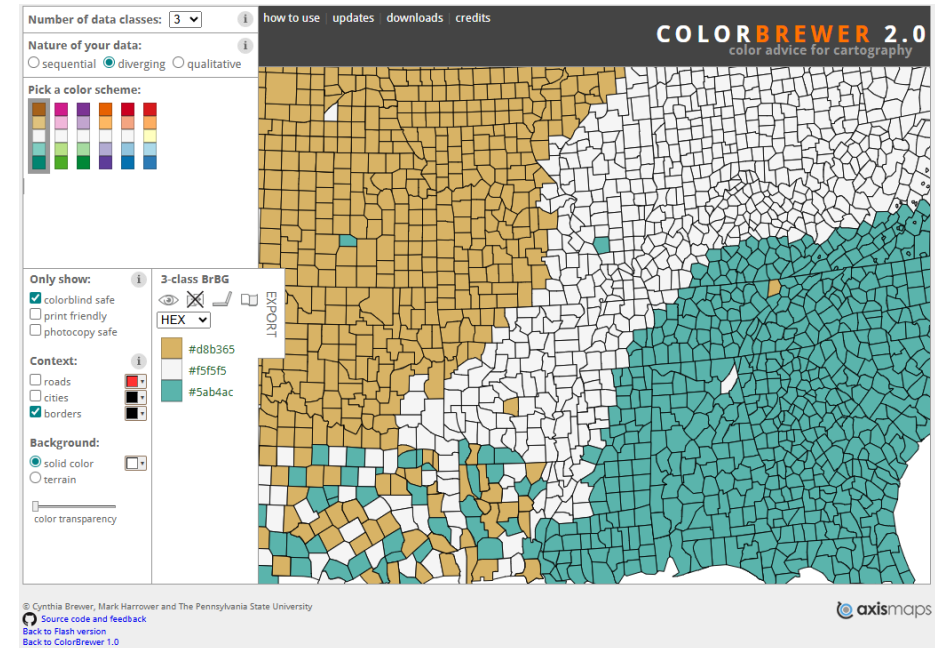
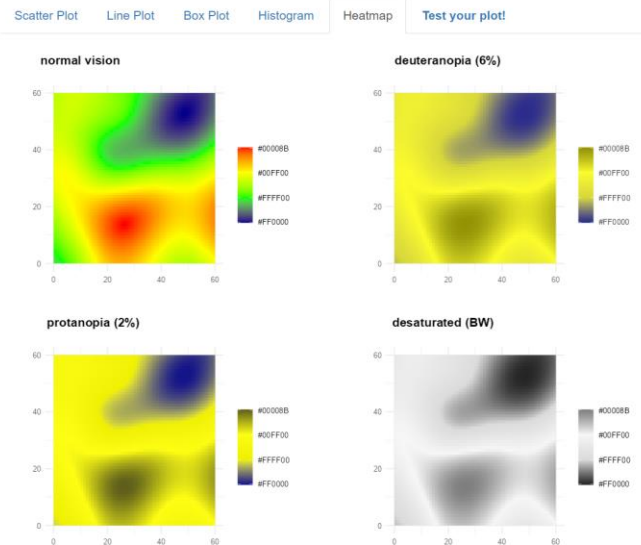
Color 2:
#FFFF00

Color 3:
#00FF00

Color 4:
#00008B

Contact me:

If you have any suggestions, questions, or need assistance, please feel free to write me at: luca.netti@glasgow.ac.uk.



Today's Plan

- Ultimate goals
 - Get an understanding of the behind the scene of statistics
 - Build intuition around probabilities
 - Learn the importance of raw data exploration
 - Learn the importance of uncertainty
- Path
 - Descriptive statistics refresher
 - First steps into statistical inference: point and interval estimates (e.g., confidence intervals)
 - Behind the scene: quantifying (un)certainty thanks to the central limit theorem and the normal distribution
 - Building intuition around probabilities with the binomial distribution
 - Behind the scene: where the numbers are coming from... the Z distribution

Descriptive Statistics Refresher

Why Do We Do Statistics?

- **Descriptive statistics:** describing the data
 - With numbers and other characteristics
 - With visuals

Descriptive Statistics Refresher

- Describing the sample data with summary statistics

With X the ordered list of x_i , the observation values, and n the sample size...

Statistics	Formula	Definition	Characteristics
Mean Or average	$\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$	Central tendency statistics , calculated as the sum of the values divided by the number of values	Is very sensitive to extreme values
Standard deviation	$sd = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$	Dispersion statistics , calculated as the mean distance of the values from the sample mean	Does not capture asymmetry
Median Or 2 nd quartile Or 50% quantile	$Med(X) = \begin{cases} X\left[\frac{n+1}{2}\right] & \text{if } n \text{ is odd} \\ \frac{X\left[\frac{n}{2}\right] + X\left[\frac{n}{2} + 1\right]}{2} & \text{if } n \text{ is even} \end{cases}$	Central tendency statistics , calculated as the middle value in a dataset when all values are arranged in order from smallest to largest	Is less sensitive to extreme values Represent the “typical observation”
Quartiles Or 1 st and 3 rd quartiles Or 25% and 75% quantiles	You can look on Google (you’ll likely never use it anyway)	The interquartile range is a dispersion statistics , calculated as the mean distance of the values from the sample mean	Captures asymmetry

Let's Try...

Vector	Mean	Standard deviation	Median	Quartiles
1, 2, 3, 4, 5	3	± 1.4	3	[2; 4]
5, 2, 1, 4, 3	3	± 1.4	3	[2; 4]
3, 3, 3, 3, 3	3	0	3	[3; 3]

- Want to train?
`sample.int(10, replace = T)`
- Check

Program	Mean	Standard deviation	Median	Quartiles
R	<code>mean()</code>	<code>sd()</code>	<code>median()</code>	<code>quantile()</code>
Excel	<code>=AVERAGE()</code>	<code>=STDEV.P()</code>	<code>=MEDIAN()</code>	<code>=QUARTILE()</code>

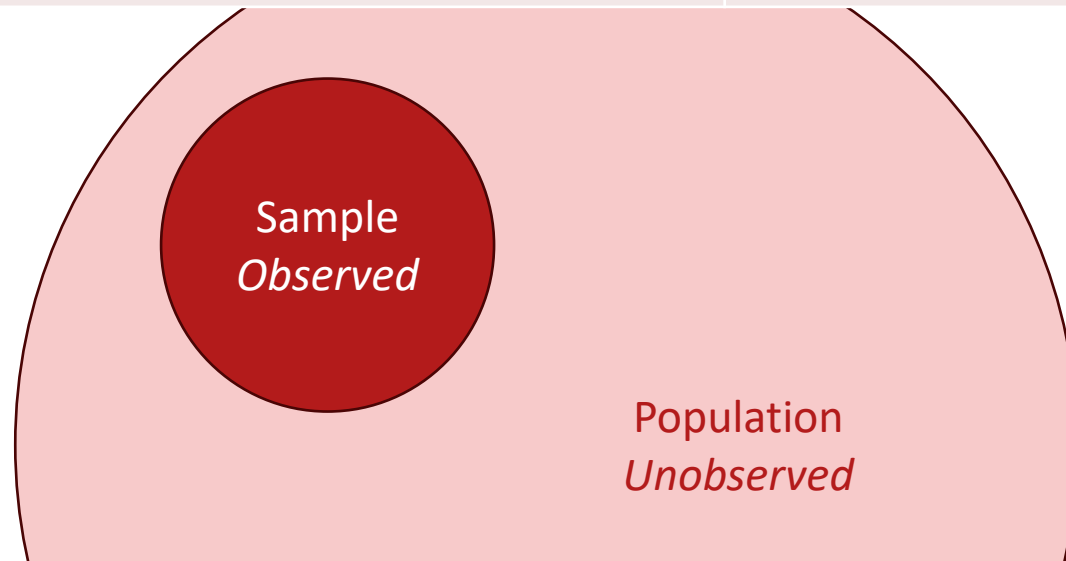
Take home: always report a central tendencies with measures of dispersion

Side Note about Samples vs Populations

- Infer the population data with summary statistics

With X the ordered list of x_i , the observation values, and n the sample size...

Statistics	Formula	Definition	Characteristics
Mean Or average	$\mu = \sum_{i=1}^n \frac{x_i}{n}$	Central tendency statistics , calculated as the sum of the values divided by the number of values	Is very sensitive to extreme values
Standard deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$	Dispersion statistics , calculated as the mean distance of the values from the sample mean	Does not capture asymmetry



Descriptive Statistics Refresher

Why do we use summary statistics?

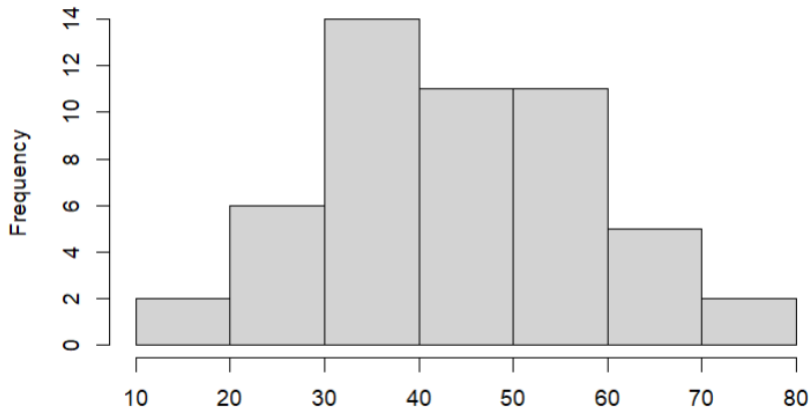
- Describe the distribution
 - **Central tendency** → *Where is the data centered?*
 - **Dispersion** → *How spread out is the data?*
- Easier to report, compare, etc... than a full description of the distribution
- But...

Take home: always report a central tendencies with measures of dispersion

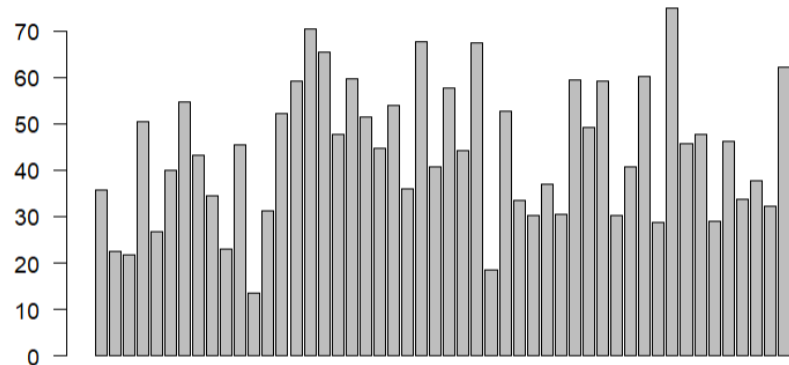
Descriptive Statistics Refresher

- Visualizing data

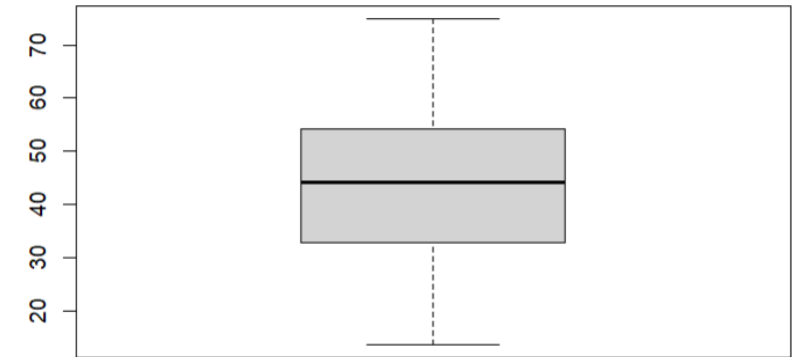
- Histogram with `hist()`
Distribution



- Barplot with `barplot()`
Individual point



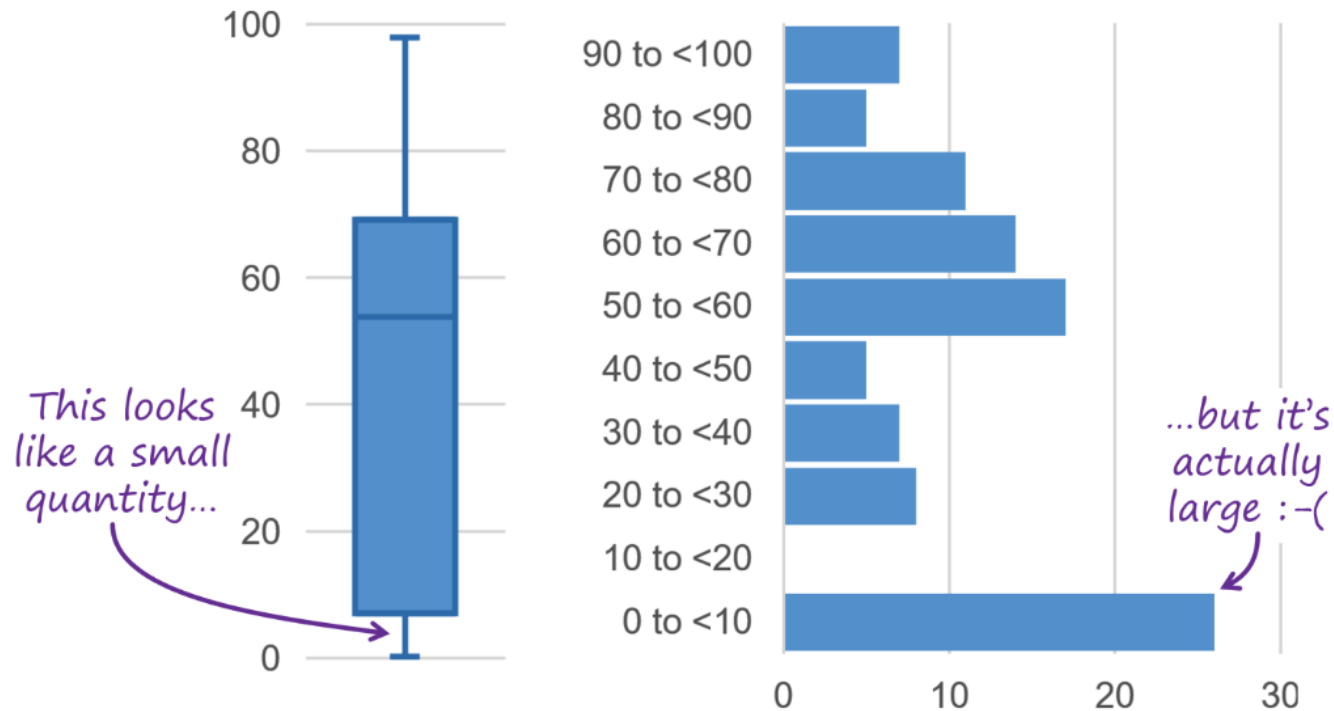
- Boxplot with `boxplot()`
Summary statistics



Which one is more informative?

Descriptive Statistics Refresher

- Visualizing data



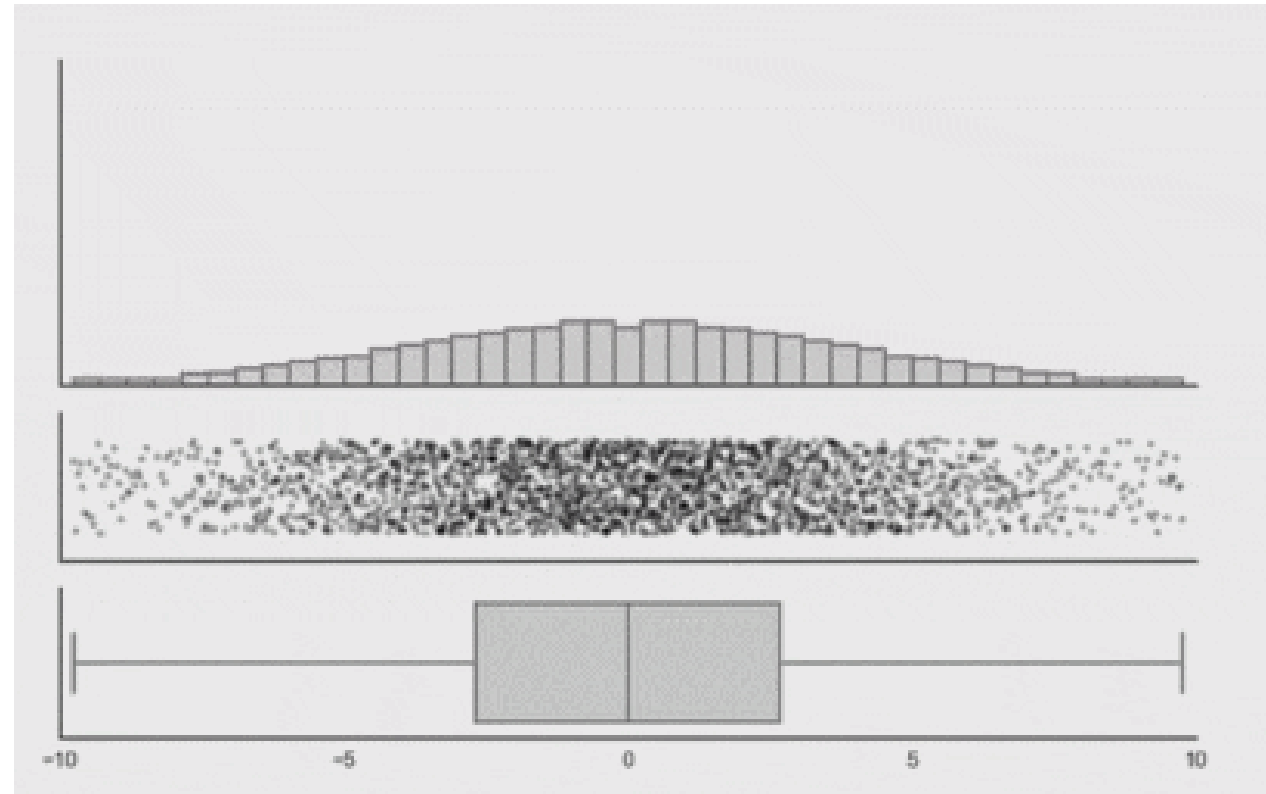
On the Parsimonious Use of Data Summary

Information

Histogram

Jittered
scatter plot

Boxplot
Summary statistics

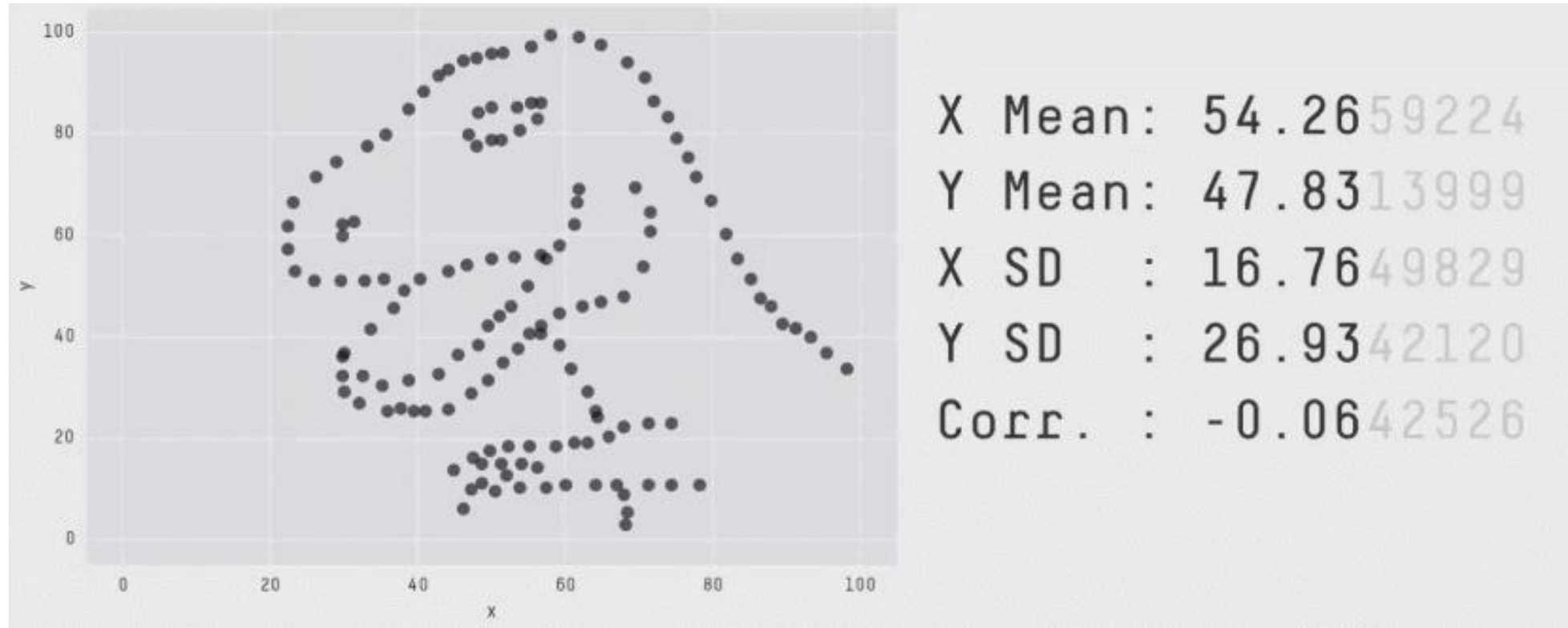


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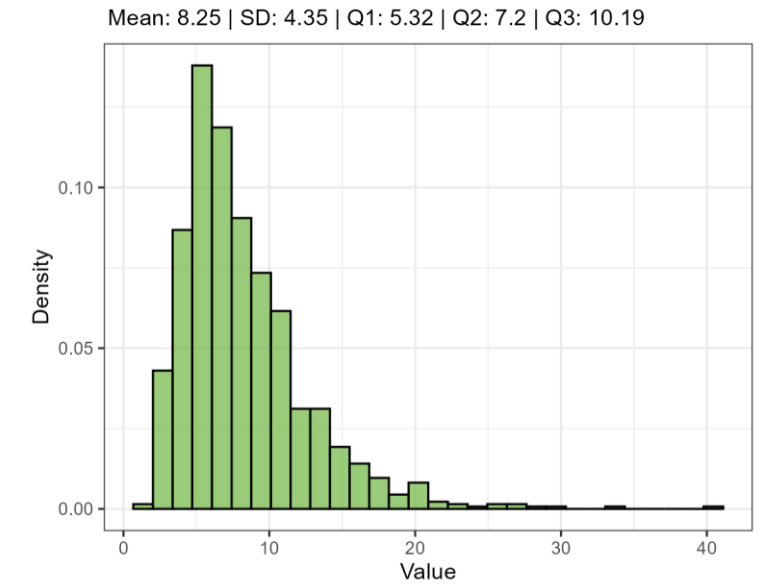
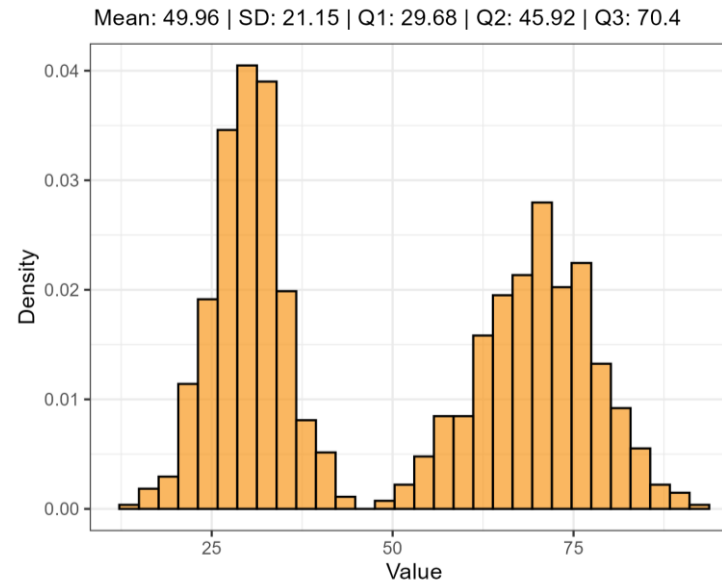
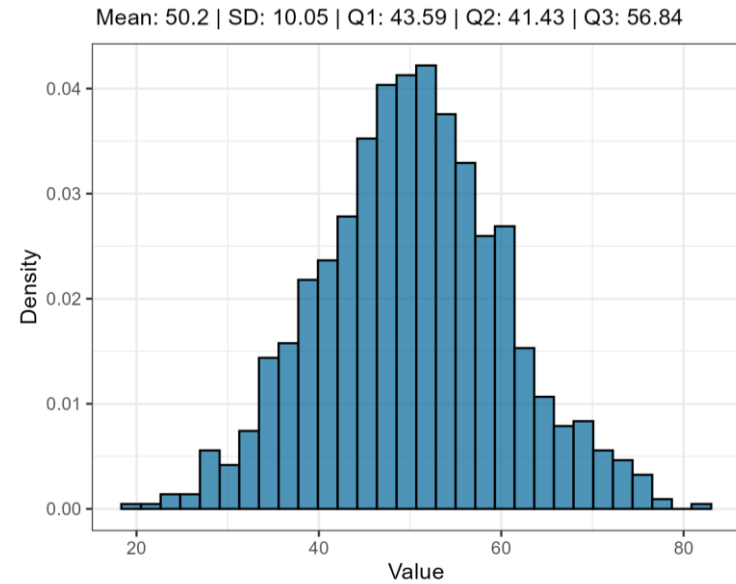
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On the Parsimonious Use of Data Summary



Always accompany summary statistics with some visualization of the raw data

On the Appropriate Use of Data Summary



- Think about how well the picked summary statistics describes the distribution of the data (symmetrical vs skewed, uni- vs multimodal...)

Pick the right summary statistics for your data (based on their distribution)

Data distribution: what is it, and why do we care?

Data distribution

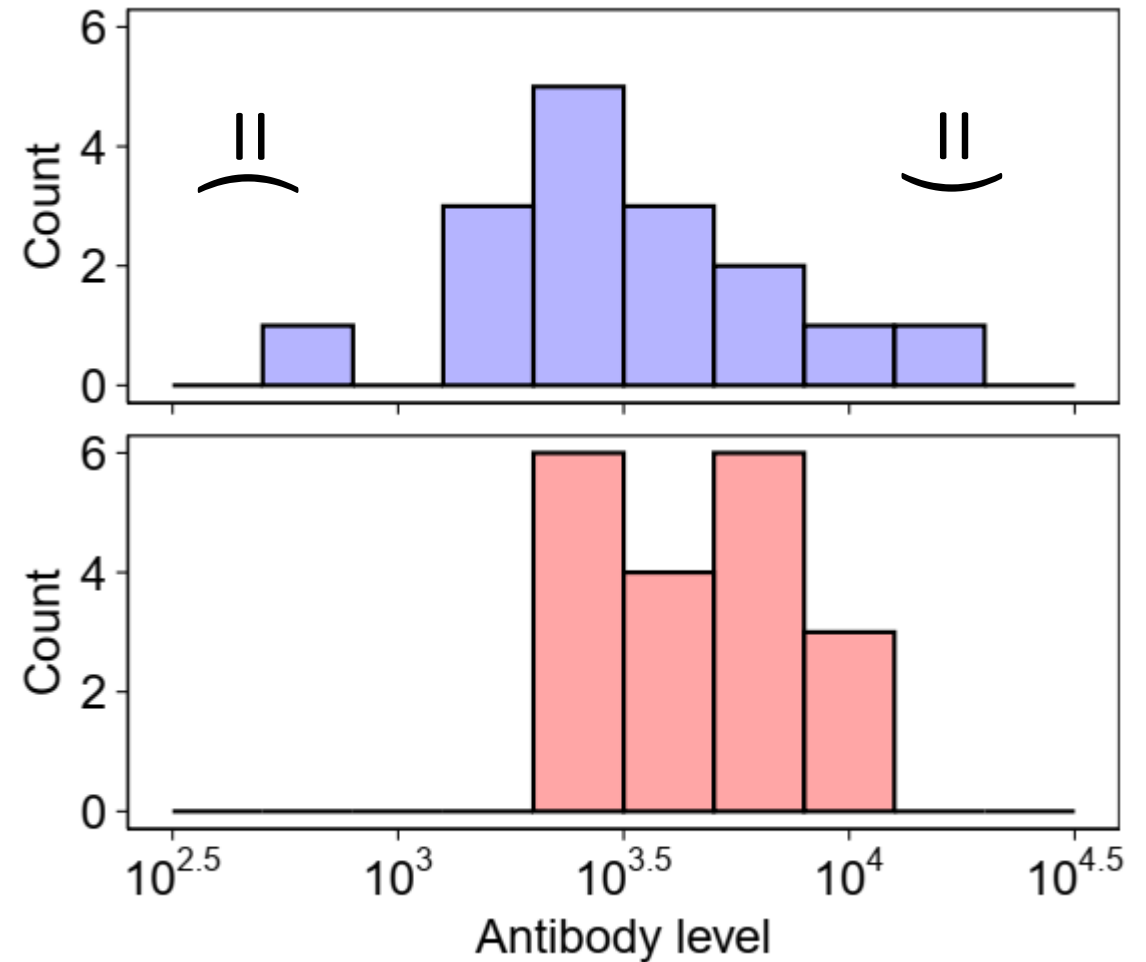
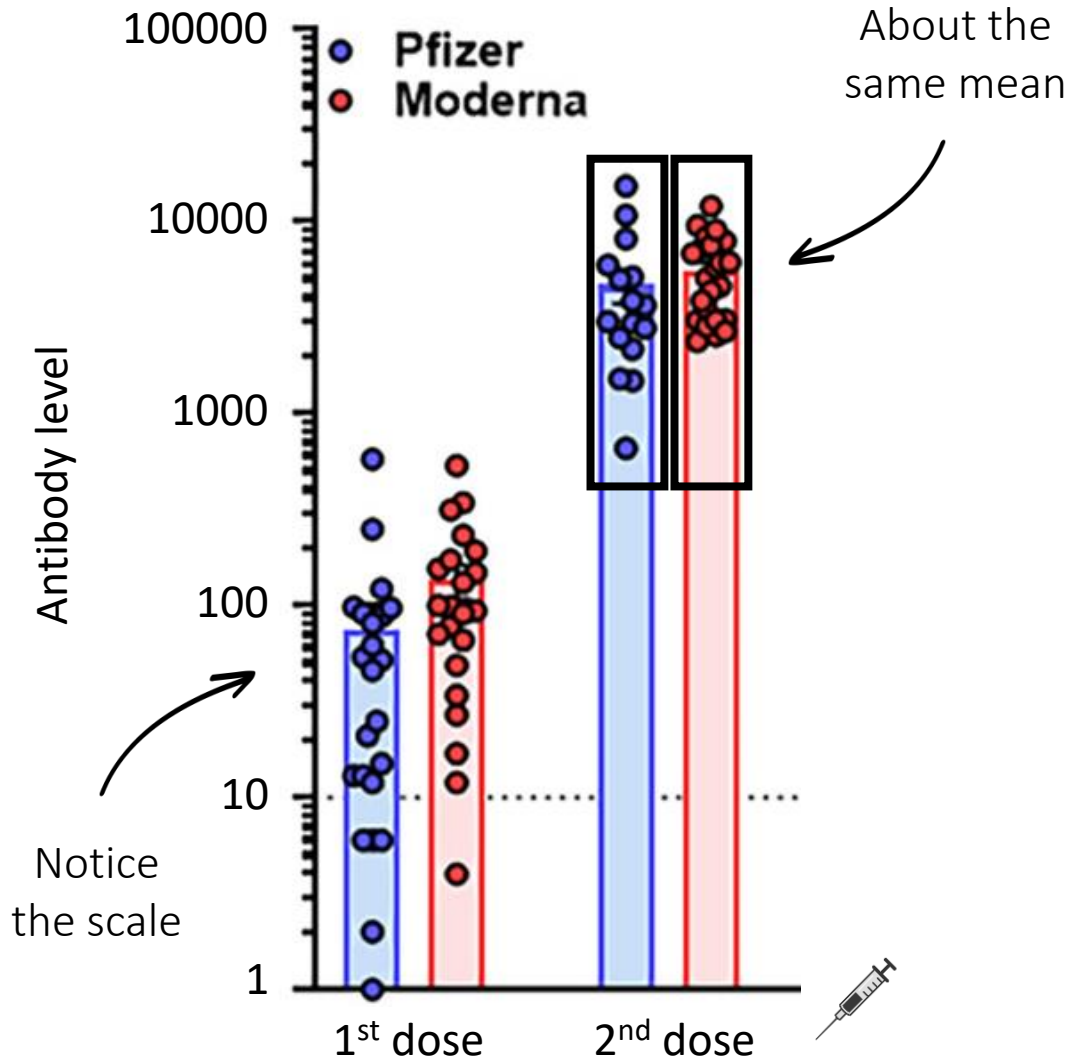
Description of all the possible values for a given variable, and how often they occur in a given data set

Describes the data

Universal questions

- What value to expect?
- Does this value differs across risk factors, treatments, etc...?

Data distribution: what is it, and why do we care?



First Steps into Statistical Inference

Point and Interval Estimates (e.g., Confidence Intervals)

Now seat comfortably and relax...

Why Do We Do Statistics?

- **Descriptive statistics:** describing the data
 - With numbers and other characteristics
 - With visuals
- **Statistical inference:** drawing conclusions
 - Usually, conclusion on a population, based on a sample
 - Can consist in **estimating numbers for one variable** (e.g., life expectancy in a population)...
 - ... or for the **association between several variables** (e.g., association between life expectancy and consumed cheese in a population)
 - To then **draw conclusion on mechanisms** (e.g., how, and why does life expectancy varies between populations)

Inference = **data-informed guess**

How Do We Do Statistics?

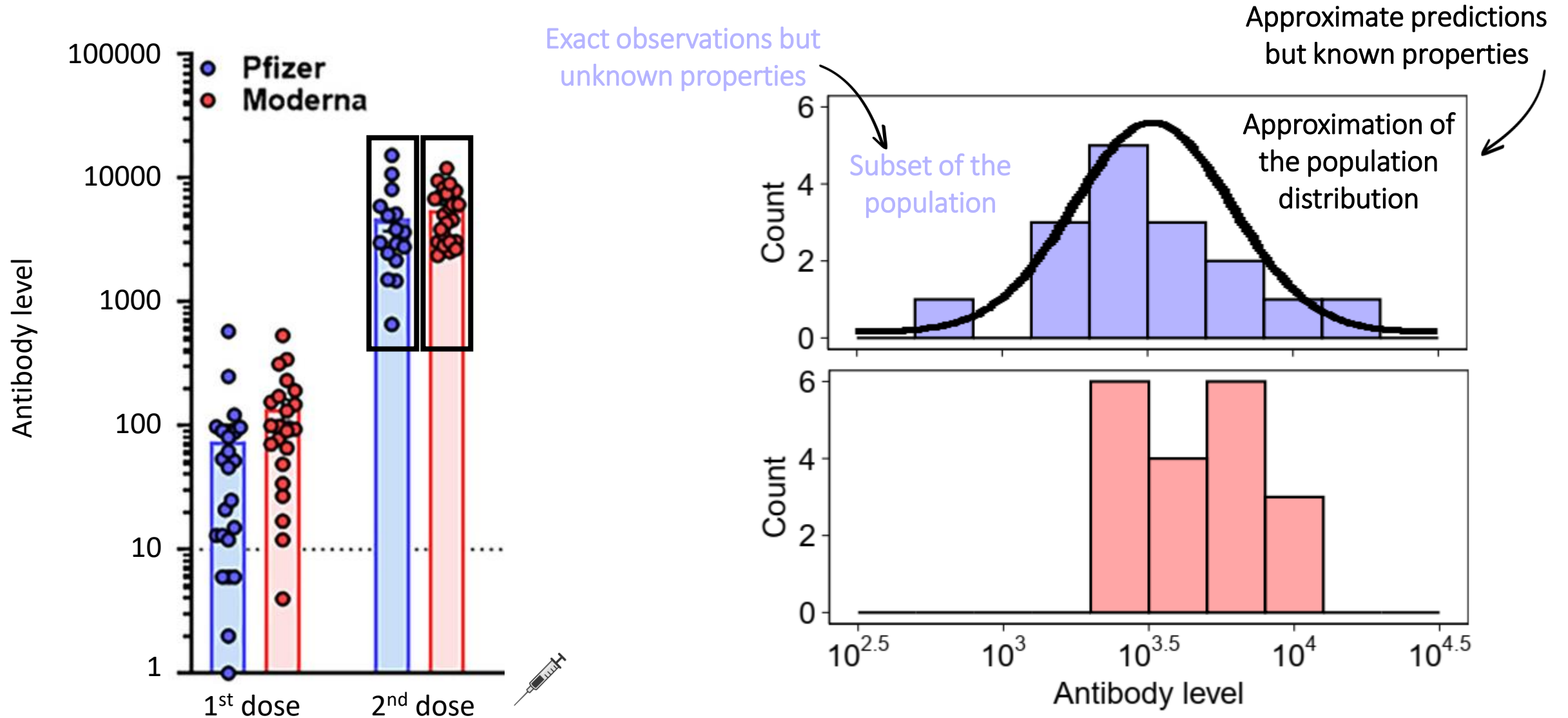
- We draw on **probabilities**
- **Probability** → *Given known parameters, what outcomes can we expect?*
- **Statistics** → *Given observed outcomes, what can we infer about unknown parameters?*

Inference = data-informed guess

UNPREDICTIBLE

FUTURE

Probability Distributions: Why Do We Care?



Probability Distributions: Why Do We Care?

Probability distribution

Mathematical function that gives the probabilities of occurrence of different possible outcomes for an experiment



tainty

Prediction is limited to probabilities

Approximates and (kinda) predicts the data

Universal question

- How to interpolate missing data?
- How to predict trends?

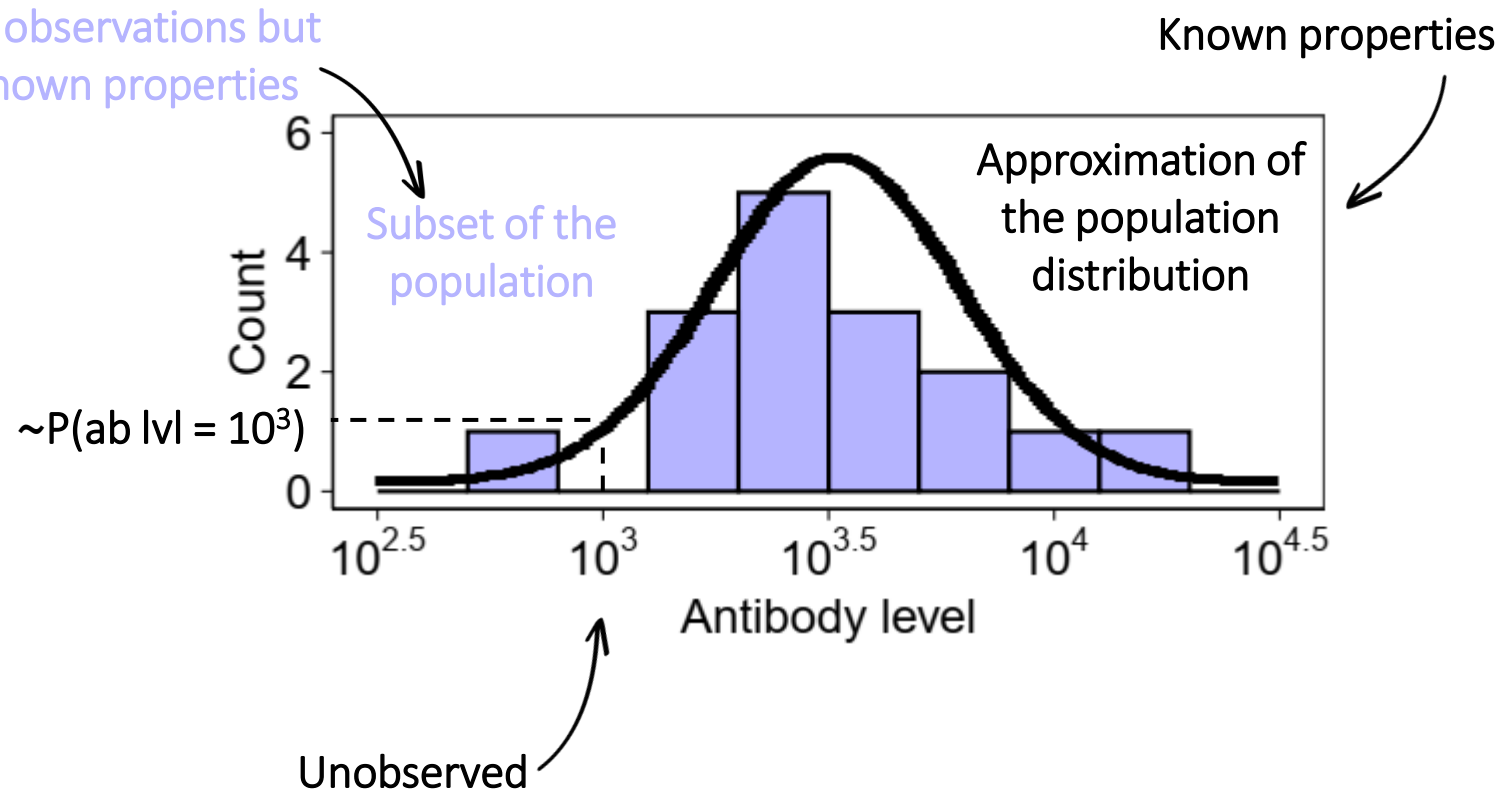
Frequency Distribution vs Probability Distribution

- **Frequency**

A measure of how often events occur, i.e., the **exact number of times each value occurs**

- **Probability**

A measure of how likely events are to occur, i.e., the **probability of occurrence of each value**

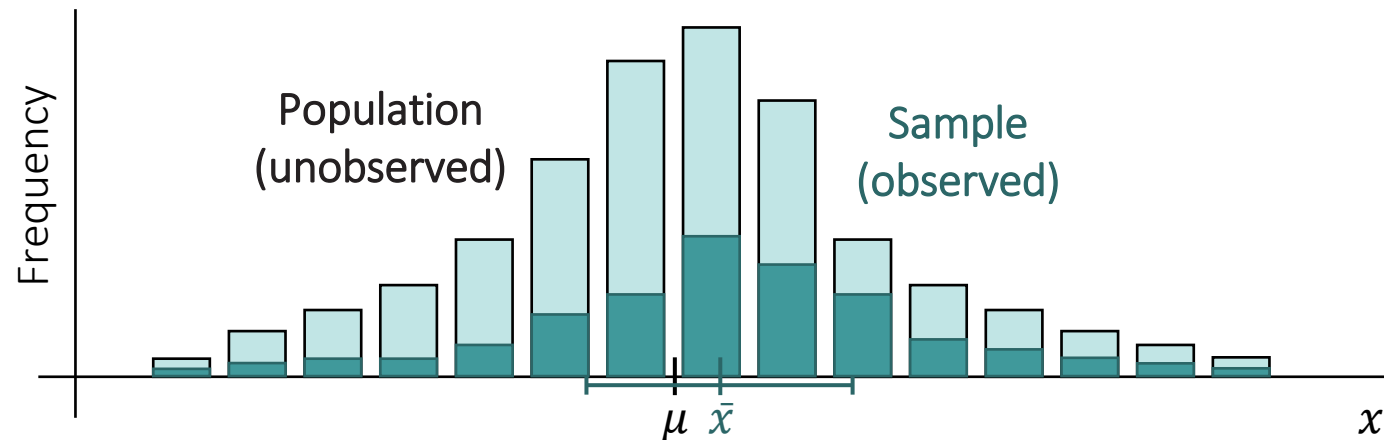


Estimates

- In statistics, **estimation** refers to the process by which one makes **inferences** about a population, based on information obtained from a sample

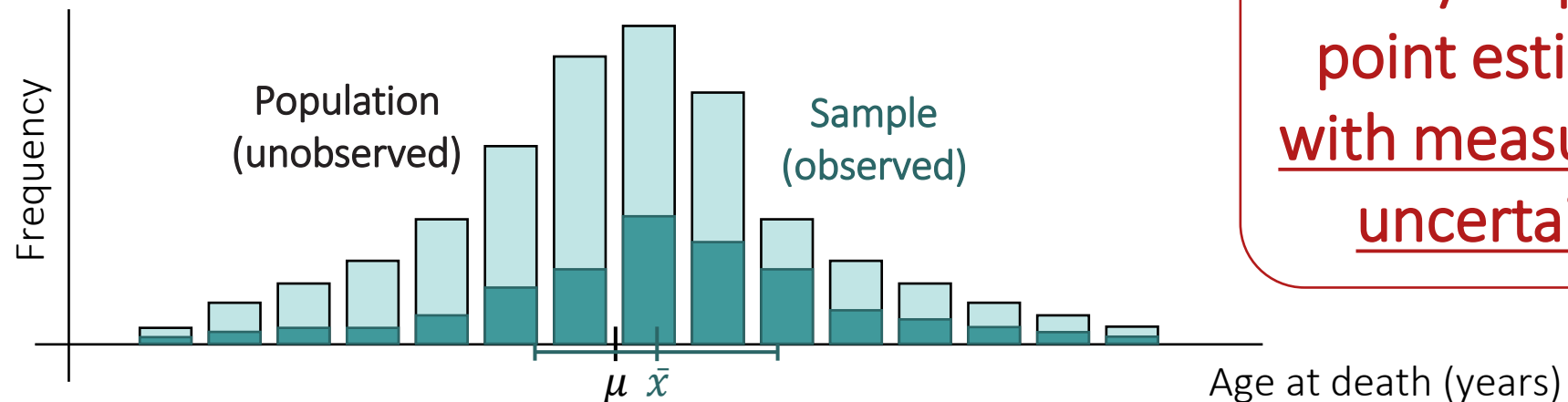
We **estimate parameters** based the **observed data**

- **Point estimation** – using the sample data to calculate a single number to estimate the parameter of interest
E.g. use the sample mean $\bar{x}$ to estimate the population mean μ (but we do not know how close the $\bar{x}$ is to μ)
- **Interval estimation** – calculate a range of reasonable values that are intended to contain the parameter of interest with a certain degree of confidence



Estimates

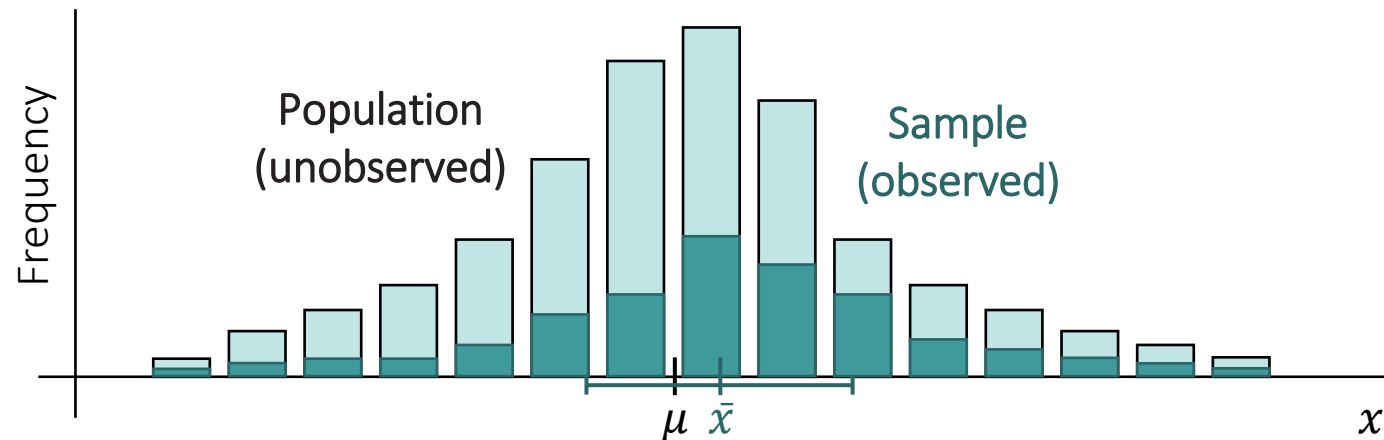
- To describe a sample, always report a **central tendency** measure with a **dispersion** measure
E.g., in the sample, average ($\pm$ standard deviation) age at death is 78.3 ± 4.5 years
Or, in the sample, median [1^{st} – 2^{nd} quartiles] age at death is 77.1 [68.2 – 83.9] years
- When inferring population parameters, always report a **point estimate** with a measure of uncertainty **as an interval**
E.g., in the population, average [95% confidence interval] age at death is 78.3 [69.1 – 87.5] years



**Always report a
point estimate
with measures of
uncertainty**

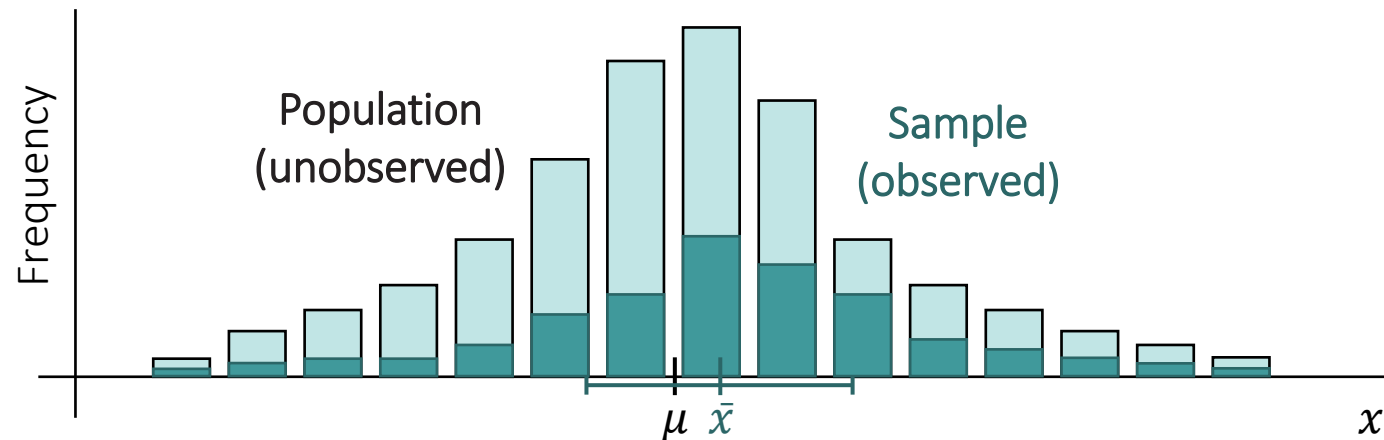
Estimates

- Spoiler: getting confidence intervals is 1 line of code in R or Excel...
- Disclaimer: this is not about learning how to calculate confidence intervals (you can ask AI), this is about understanding the concepts behind to correctly interpret and/or use them



Estimates

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$... and thanks to the central limit theorem, we can infer μ



Behind the Scene: Quantifying (Un)certainty

Thanks to the Central Limit Theorem and the Normal Distribution

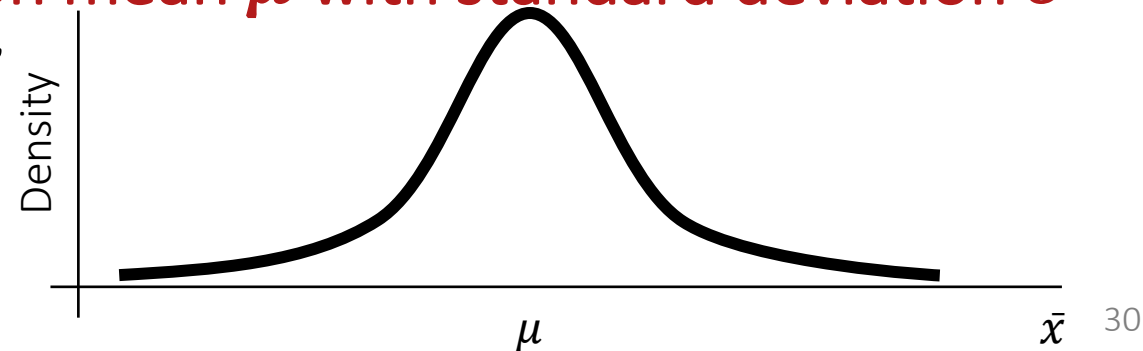
The Central Limit Theorem to the Rescue

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$
- **Central limit theorem**: given a population of any (non-normal) functional form with a mean μ and standard deviation σ , the sampling distribution of $\bar{x}$, computed from samples of size n from this population, will have mean μ and standard deviation $\sigma/\sqrt{n}$ and will be approximately **normally distributed** when the sample size is large

$$\bar{x} \sim N(\mu, \sigma/\sqrt{n})$$

→ **For n large enough, the distribution of the sample mean $\bar{x}$ is approximately normal, and centered around the population mean μ with standard deviation σ**

- Rule of thumb: $n \geq 30$ considered “large”
- Side note: variance = standard deviation²



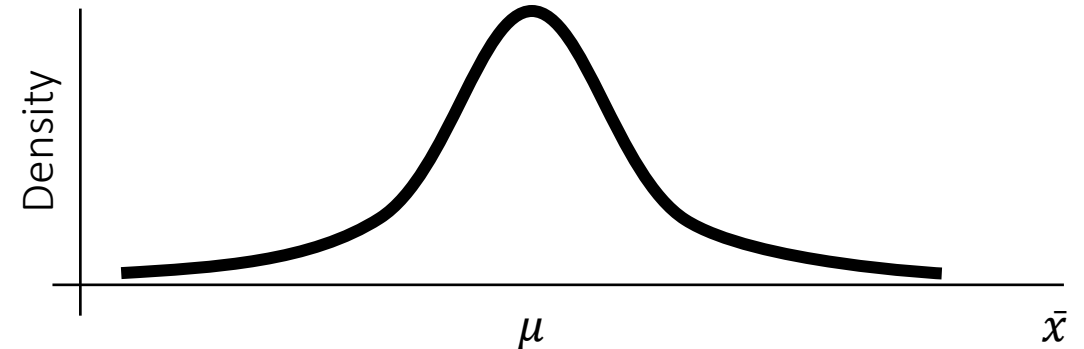
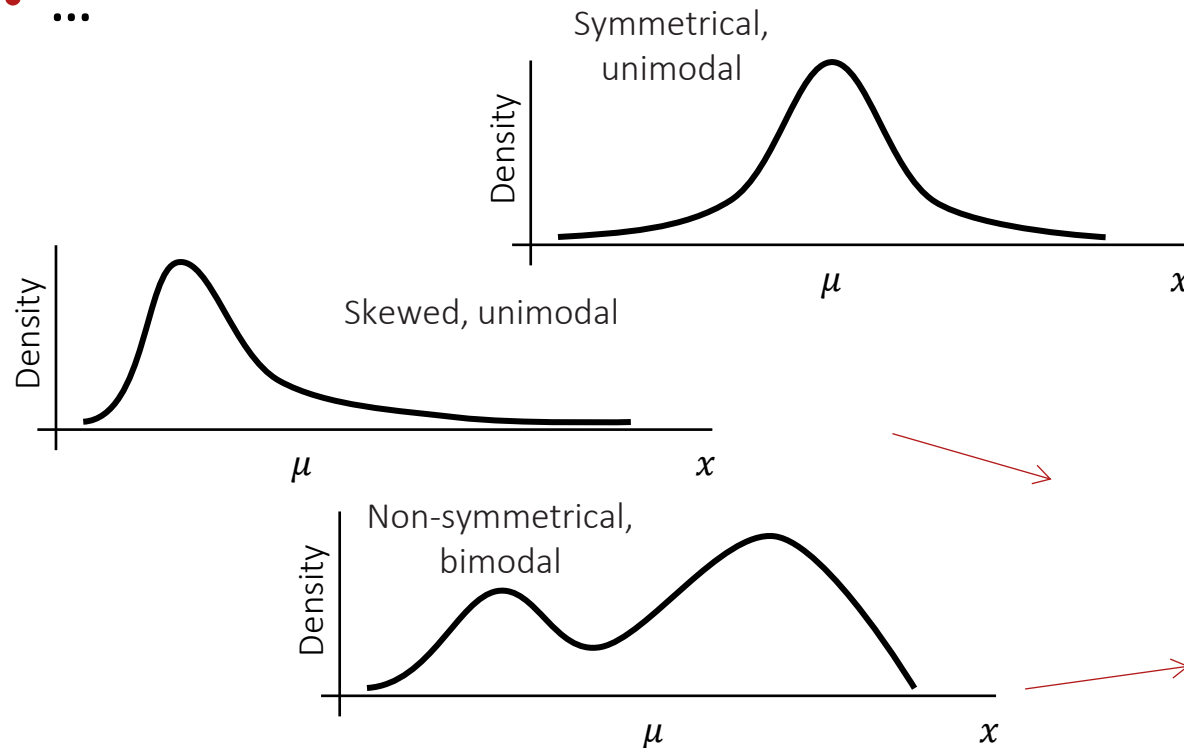
The Central Limit Theorem to the Rescue

- Population of 100,000 individuals with mean μ

- Sample #1 of 5,000 individuals with mean $\bar{x}_1$
- Sample #2 of 5,000 individuals with mean $\bar{x}_2$
- Sample #3 of 5,000 individuals with mean $\bar{x}_3$
- Sample # i of 5,000 individuals with mean $\bar{x}_i$
- ...

$\bar{x}$ follows a normal distribution centered around μ with standard deviation $\sigma/\sqrt{n}$

Independently from the distribution of x



The Central Limit Theorem to the Rescue

- If we know the sample mean, we can infer the population mean...
- ... Because the sample mean follows a normal distribution
- And why is this cool? Because we know everything about the normal distribution!

The Normal Distribution

- Use to describe continuous variables with symmetric distributions
 - Examples?
 - Size, weight... lots of biological variables

Normal distribution

$$P(x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Parameters

μ : mean

σ : standard deviation

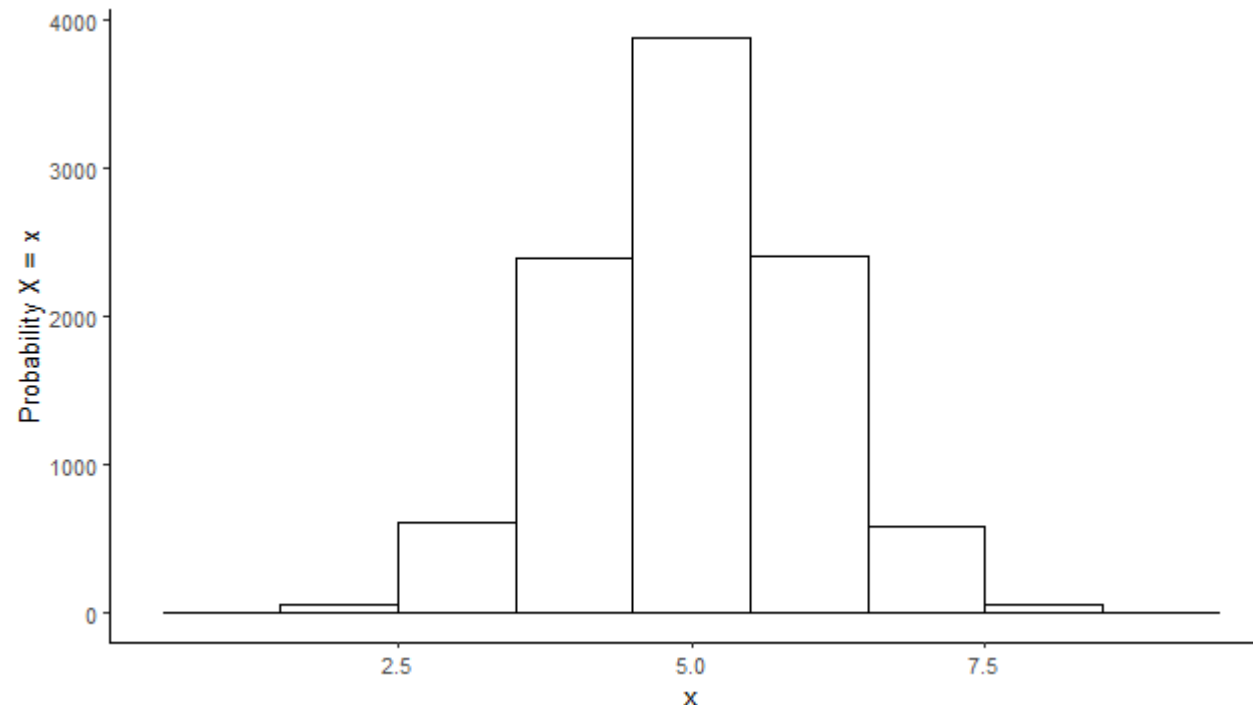
* You do not need to know this, you just need to know we (collectively as humans) know it, so we can do math with it

The Normal Distribution

- Probability density function (PDF)

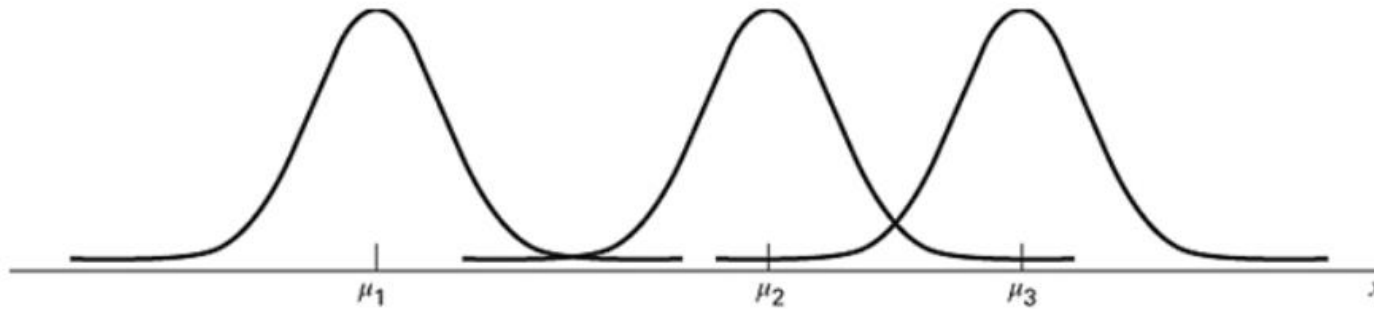
$$f(X) = P(X = x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- X is a continuous random variable
 - Expected value $E[X] = \mu$
- $f(X)$ is a probability $f(X) \in [0; 1]$
Variance $\text{Var}[X] = \sigma^2$

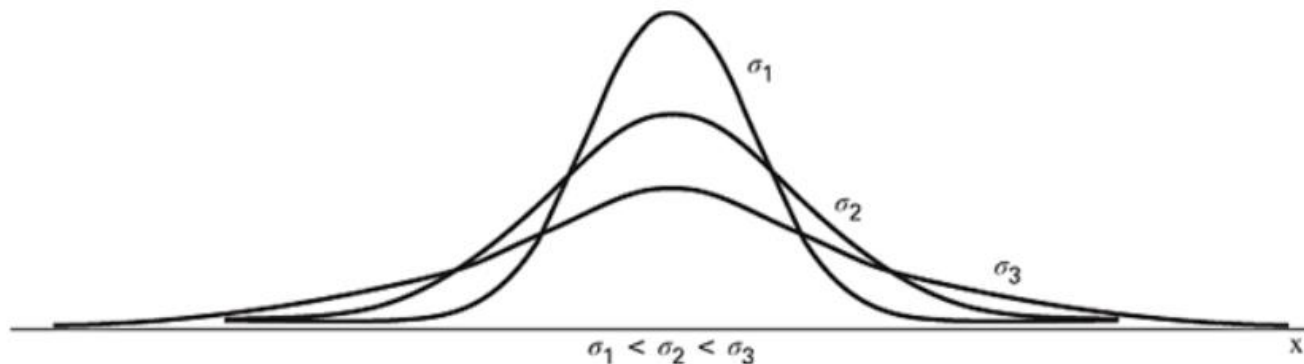


The Normal Distribution

- Use to describe continuous variables with symmetric distributions
 - The mean and variance completely describe the normal curve



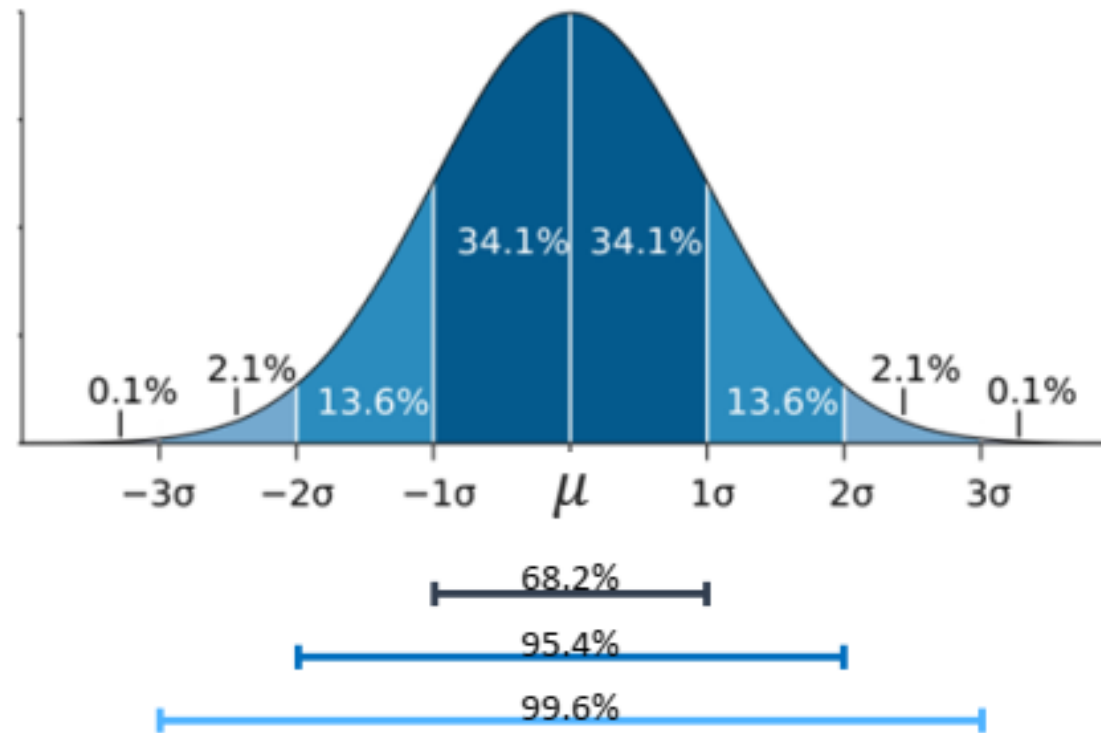
$$\mu_1 < \mu_2 < \mu_3$$



$$\sigma_1 < \sigma_2 < \sigma_3$$

The Normal Distribution

- Use to describe continuous variables with symmetric distributions
 - The mean and variance completely describe the normal curve



The Normal Distribution

- The curve is called a probability density function or the bell-shaped curve.
- Continuous random variable
- Symmetric around its mean, μ
- The mean = median = mode (unimodal)
- The total area beneath the curve sums to 1
- σ the standard deviation = the amount of dispersion around the mean

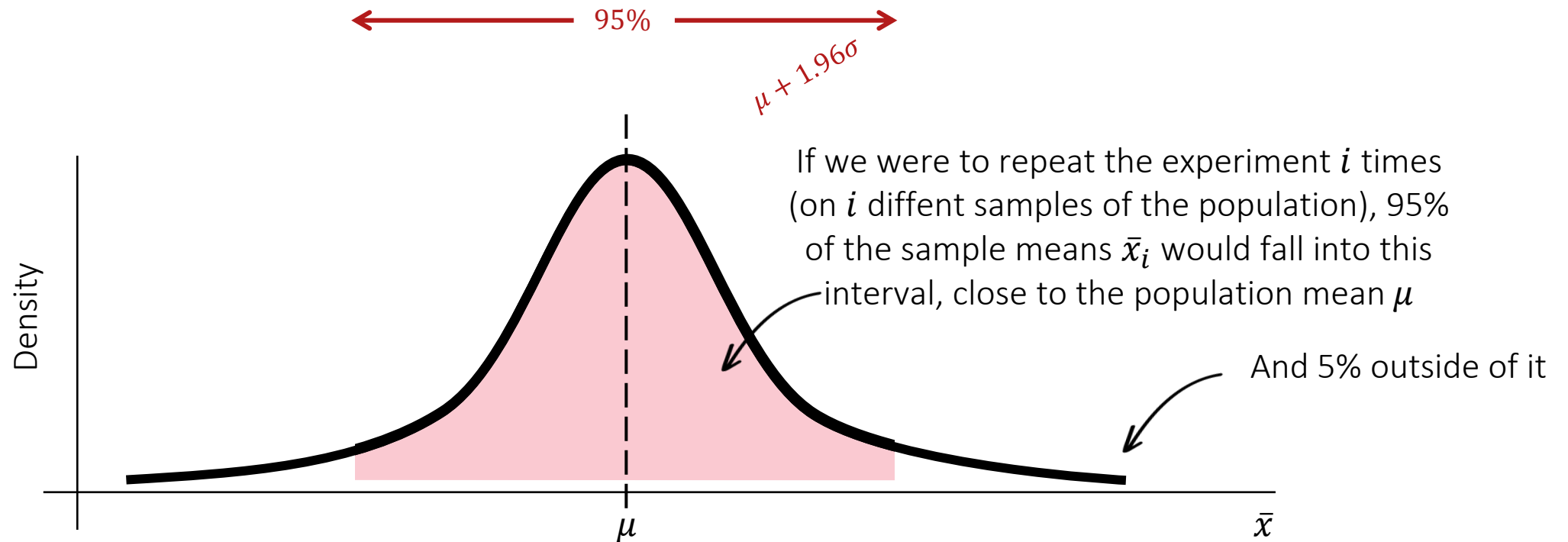
The Central Limit Theorem to the Rescue

- If we know the sample mean, we can infer the population mean...
- ... Because the sample mean follows a normal distribution
- And why is this cool? Because we know everything about the normal distribution!
- It does not mean we know exactly where the true population mean is...
- It means **we know where the true population mean likely is, $\pm$ some uncertainty...**
- ... Which we can quantify from what we know about the normal distribution

We never know the true population mean, but we know the distribution of the sample mean is centered around it, so we can infer where it likely is

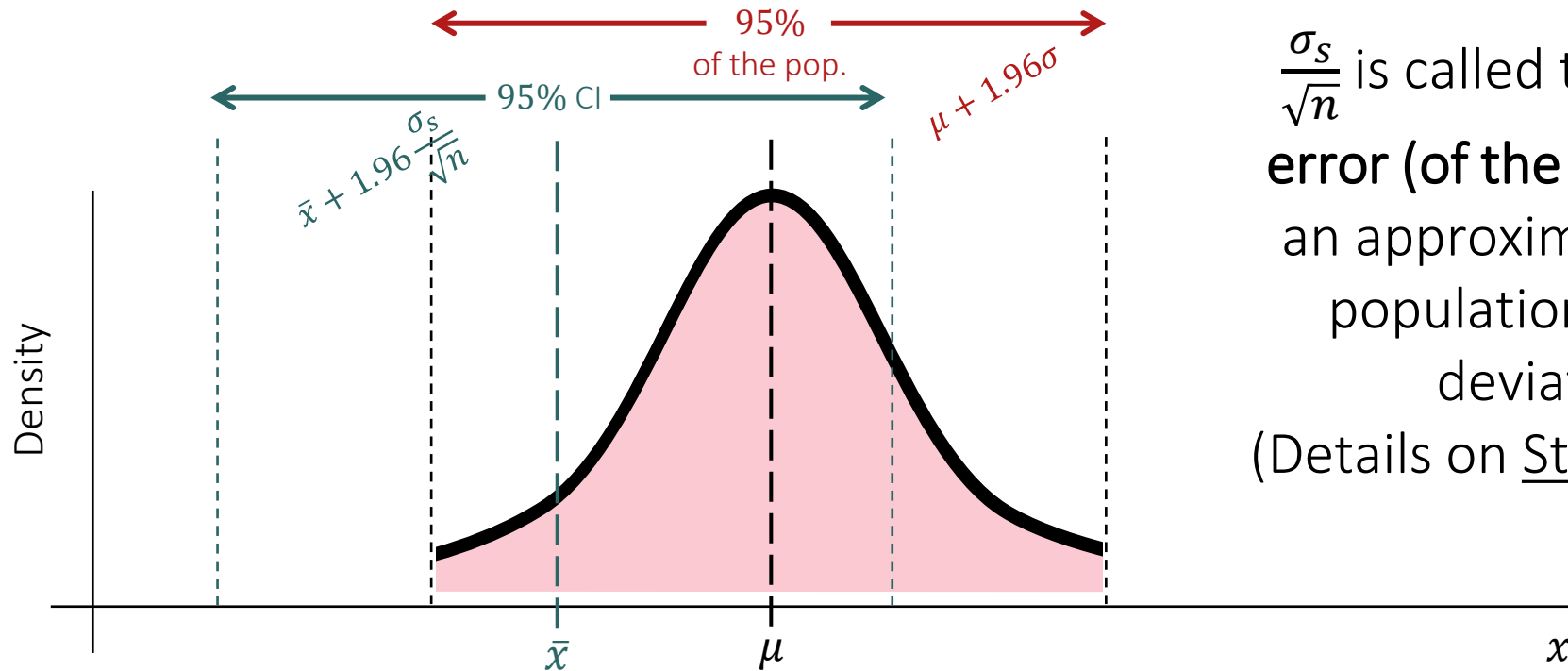
Confidence Intervals: Definition

- If we knew the population parameters...



Confidence Intervals: Definition

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

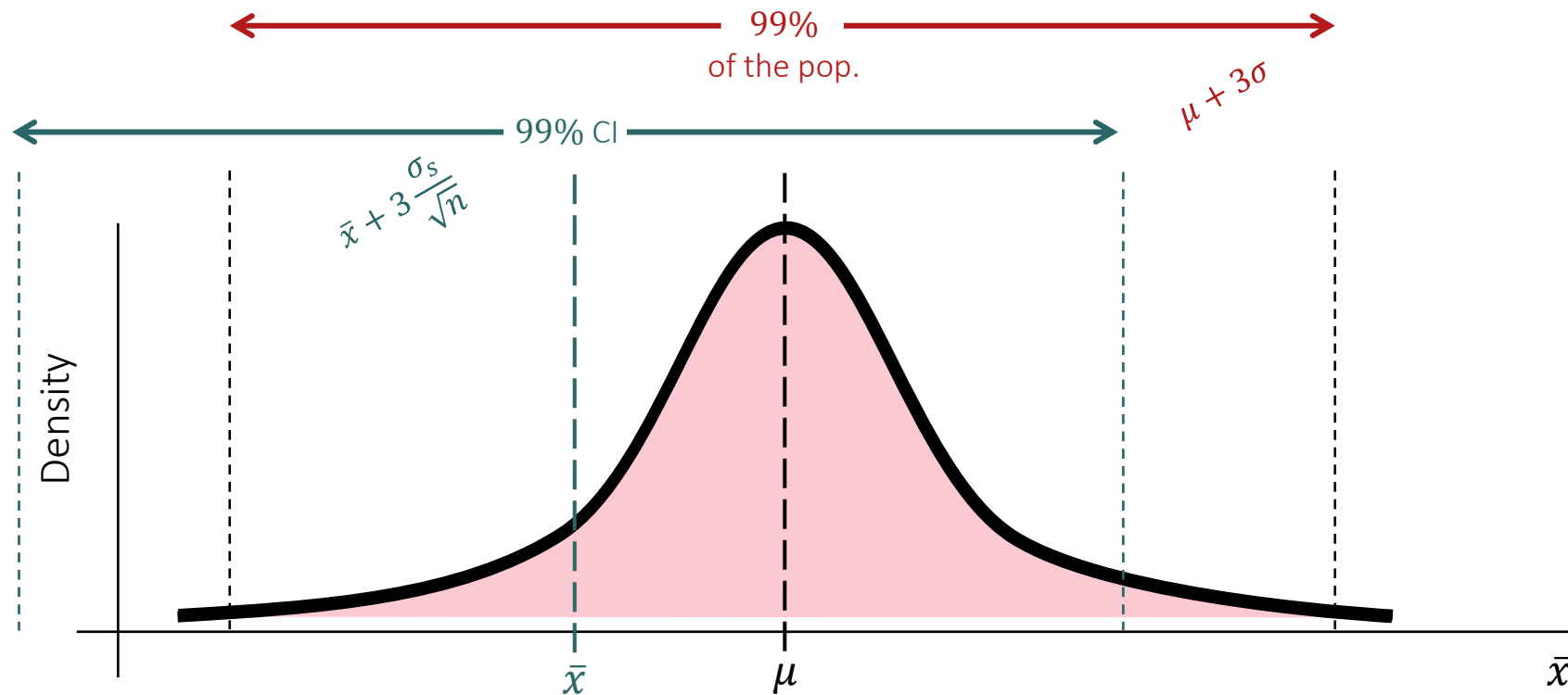


$\frac{\sigma_s}{\sqrt{n}}$ is called the **standard error (of the mean)** and is an approximation of the population standard deviation σ
(Details on [StackExchange](#))

95% CI = $[\bar{x} - 1.96 \frac{\sigma_s}{\sqrt{n}}; \bar{x} + 1.96 \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, and n the sample size

Confidence Intervals: Definition

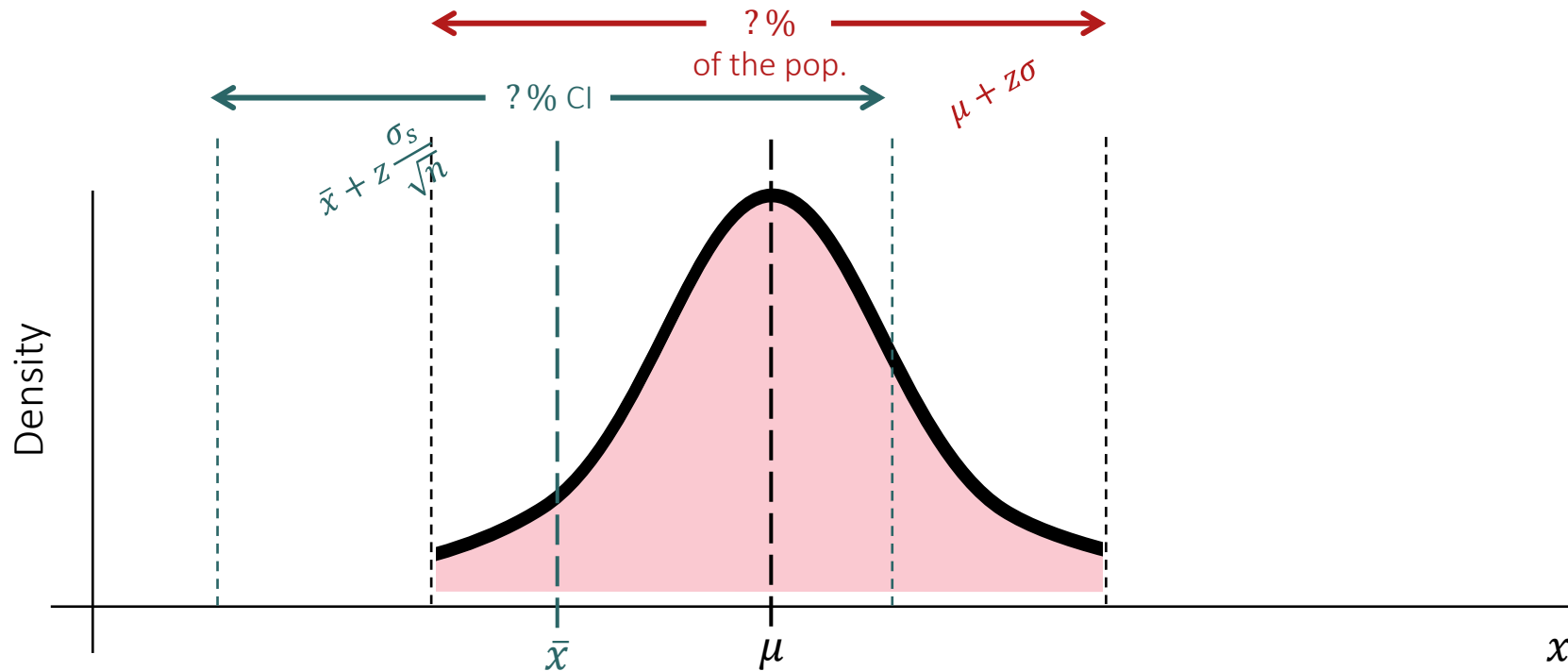
- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



99% CI = $[\bar{x} - 3 \frac{\sigma_s}{\sqrt{n}}; \bar{x} + 3 \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, and n the sample size

Confidence Intervals: Definition

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

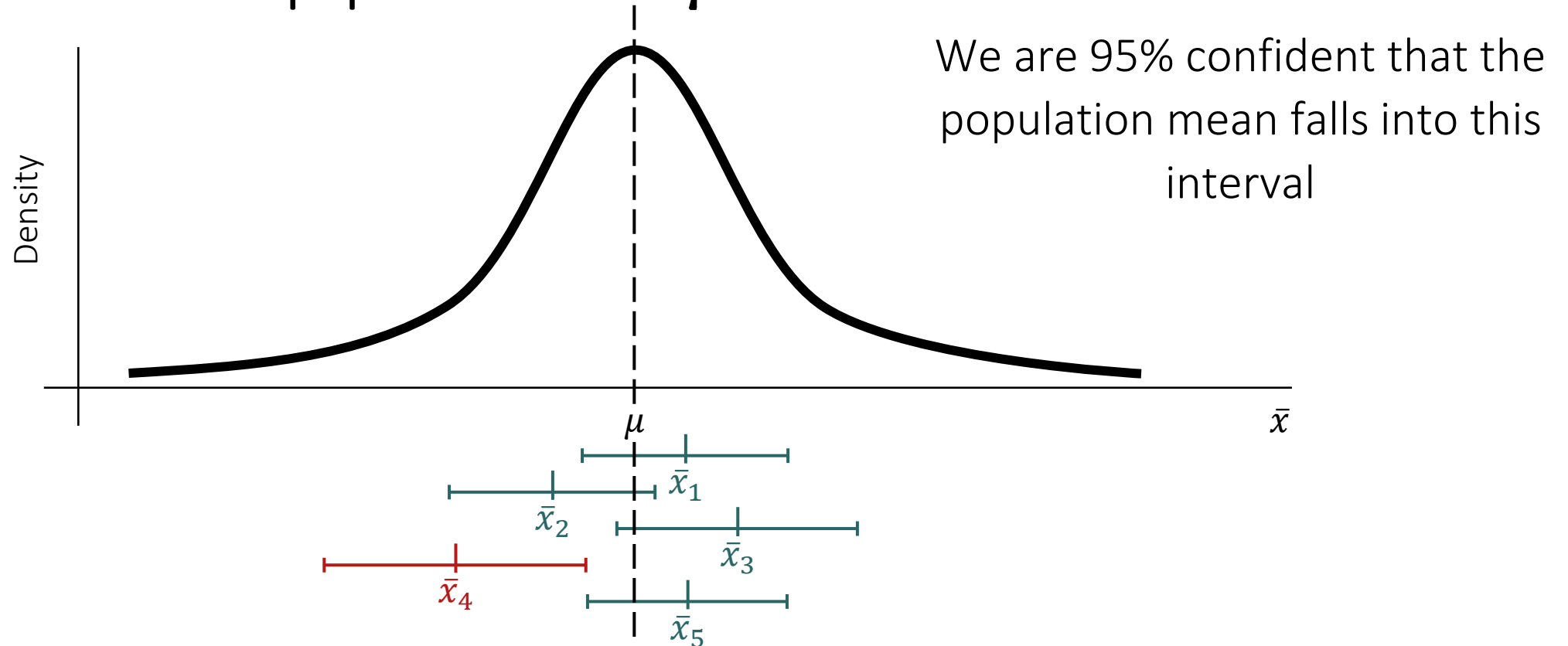


$CI = [\bar{x} - z \frac{\sigma_s}{\sqrt{n}}; \bar{x} + z \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

Confidence Intervals: Interpretation

- Starting from the sample parameters...

For 95% confidence intervals, if we were to repeat the experiment i times (on i different samples of the population), **95% of the i defined confidence intervals would contain the true population mean μ**



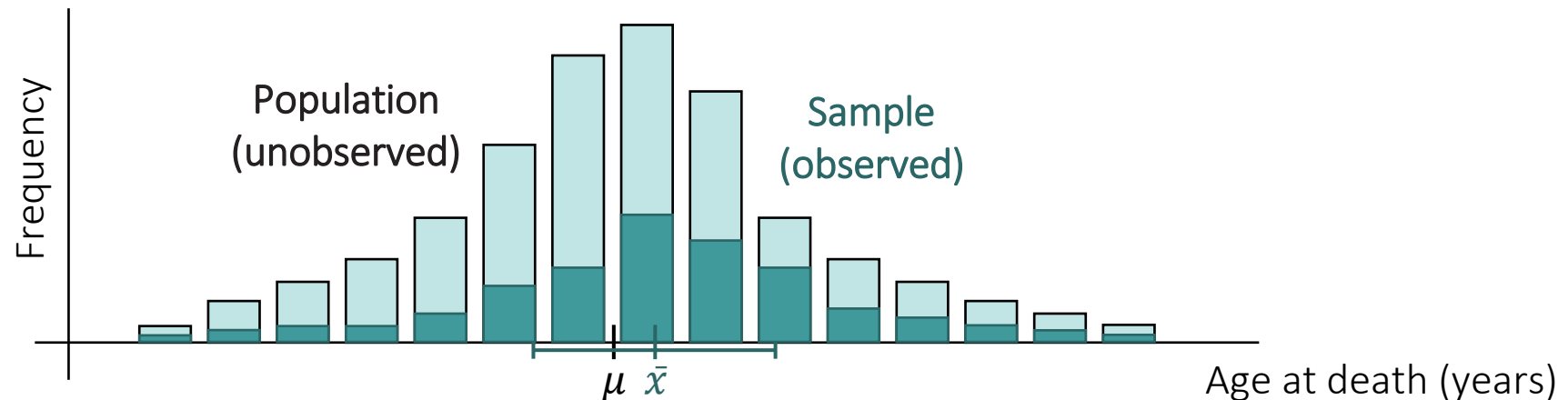
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- To describe a sample, always report a **central tendency** measure with a **dispersion** measure

E.g., in the sample, average ($\pm$ standard deviation) age at death is 78.3 ± 4.5 years
When inferring population parameters, always report a **point estimate** with a measure of uncertainty **as an interval**

E.g., in the population, average [95% confidence interval] age at death is
 $78.3 [69.1 - 87.5]$ years

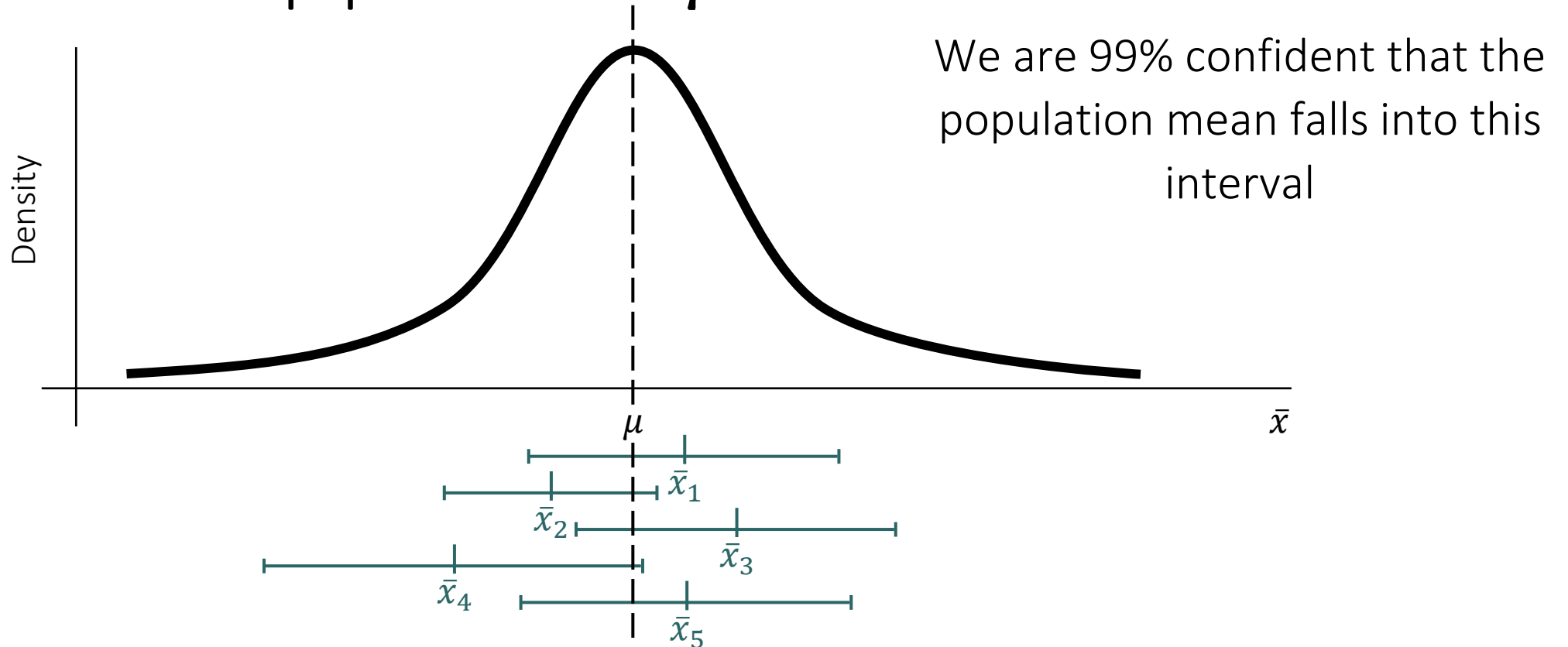
→ We are 95% confident that the average age at death in the population falls into the interval $78.3 [69.1 - 87.5]$ years



Confidence Intervals: Interpretation

- Starting from the sample parameters...

For 99% confidence intervals, if we were to repeat the experiment i times (on i different samples of the population), **99% of the i defined confidence intervals would contain the true population mean μ**



Confidence Intervals: Interpretation

- Starting from the sample parameters...

For 99% confidence intervals, if we were to repeat the experiment i times (on i different samples of the population), **99% of the i defined confidence intervals would contain the true population mean μ**

- The **wider** the confidence interval, the **more confident** we are to overlap with the population parameter, but the **less informative** the estimated parameter is
- The higher the sample size n , the tighter the confidence interval

Increasing sample size enables a higher precision with high confidence
(i.e., lower uncertainty)

Confidence Intervals: Interpretation

- The same procedure applies to means and proportions

Mean

$CI = [\bar{x} - z \frac{\sigma_s}{\sqrt{n}}; \bar{x} + z \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

E.g., age at death in a population

Proportion

$CI = [\bar{p} - z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}; \bar{p} + z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}]$
with $\bar{p}$ the sample proportion, n the sample size and z the z-score for the confidence level

E.g., number of infected people in a population (prevalence)

Confidence Intervals: Why Do We Care?

- Reporting point estimates with confidence intervals provides information on the uncertainty around the point estimates

It is a critical information to interpret numbers

Never trust a mean not presented with its confidence interval!

- It is the first step in **hypothesis testing**

Most hypothesis testing procedures are based on comparisons of means

Building Intuitions around Probabilities

The Binomial Distribution

The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other
 - Examples?
 - Coin flip
 - The number of heads we obtain (**x**) is based on the probability of flipping a head (**p**) and the number of times we toss the coin (**n**)

$$P(x \text{ heads}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- $P(2 \text{ heads})$ when tossing the coin 3 times
 - $x = 2; n = 3$
 - $p = 0.5$
 - The probability of getting 2 heads among 3 tosses =

probability of getting 2 heads AND 3-1 = 1 tails

in every possible order (combinations without replacement)

$$= 0.375$$

$$p^x (1 - p)^{n-x}$$

$$\binom{n}{x} = \frac{n!}{(n-x)!x!}$$

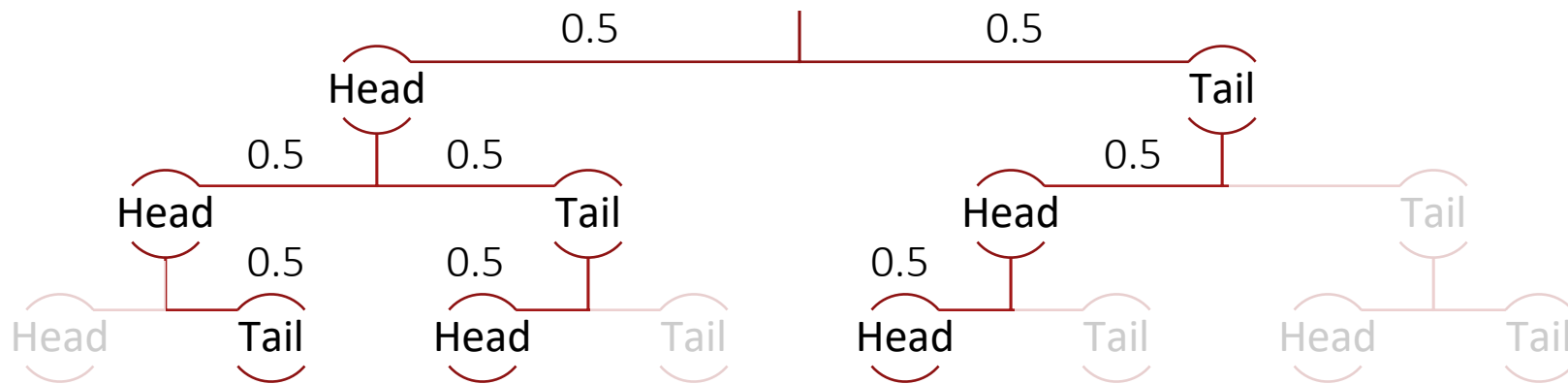
$$0.5^2 (1 - 0.5)^{3-2}$$

$$\binom{3}{2} = \frac{3 \times 2 \times 1}{(1)(2 \times 1)} = 3$$



The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other
 - Examples?
 - Coin flip, a maybe more intuitive presentation
 - $P(2 \text{ heads})$ when tossing the coin 3 times



Probability tree

What if the coin was unfair?

What if we were playing with a dice instead?

$$\begin{aligned} &= P([H \cap H \cap T] \cup [H \cap T \cap H] \cup [T \cap H \cap H]) \\ &= [P(H) \times P(H) \times P(T)] + [P(H) \times P(T) \times P(H)] + [P(T) \times P(H) \times P(H)] \\ &= 0.5^3 + 0.5^3 + 0.5^3 \\ &= 0.375 \end{aligned}$$

What if we cared about the order?

The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other

Count = (usually) discrete

Binomial distribution

Combinations

$$P(x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Failures

Successes

Parameters

n : number of trials

p : probability of success per trial

The Binomial Distribution

- Key properties of the binomial distribution
 - The outcome of a random process or experiment can result in only one of two **mutually exclusive outcomes**, such as a treatment is “success/failure”; a patient is “alive/dead” or “treated/non-treated”
 - The **outcomes are independent**, that is the outcome of any particular trial (or individual) is not affected by the outcome of any other event (or individual)
 - The **probability of a success (p) remains constant** for each trial

The Binomial Distribution

- Used to describe binary events, by counting the occurrence of one event or the other
 - Examples?
 - Likelihood of infection with rabies
 - The number of infected people (**x**) is based on the probability of successful infection after a bite (**p**) and the number of people bitten by rabid animals (**n**)

$$P(x \text{ infections}) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Modulated by
vaccine

Modulated by the
number of bites

... And the
number of rabid
animals

- Help think about mitigation measures
- Basis to compare mitigation measures



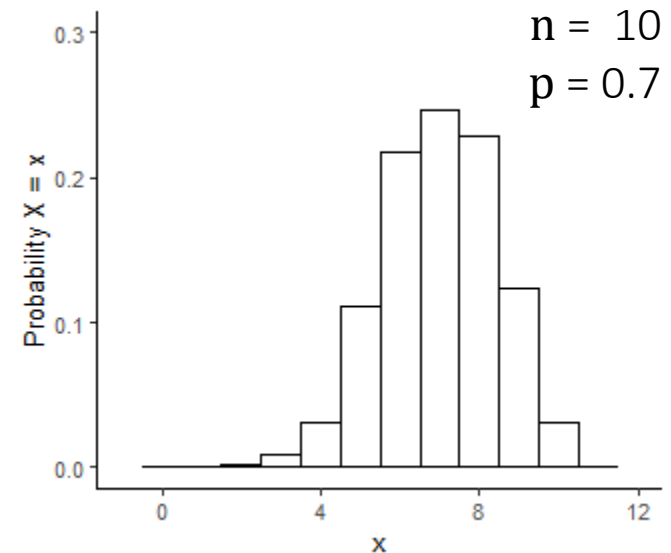
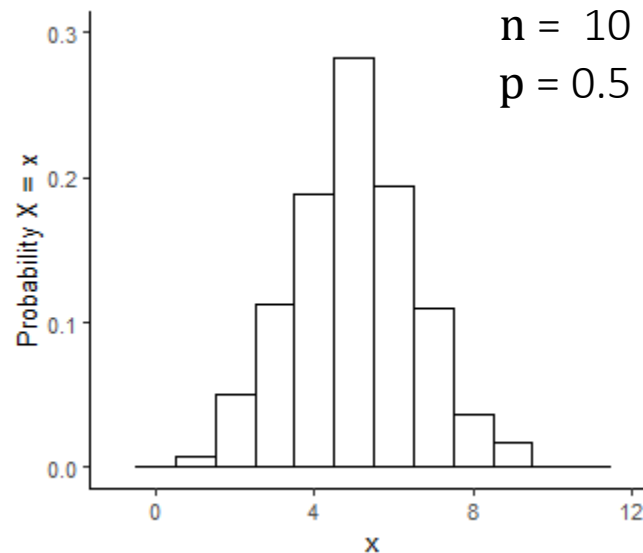
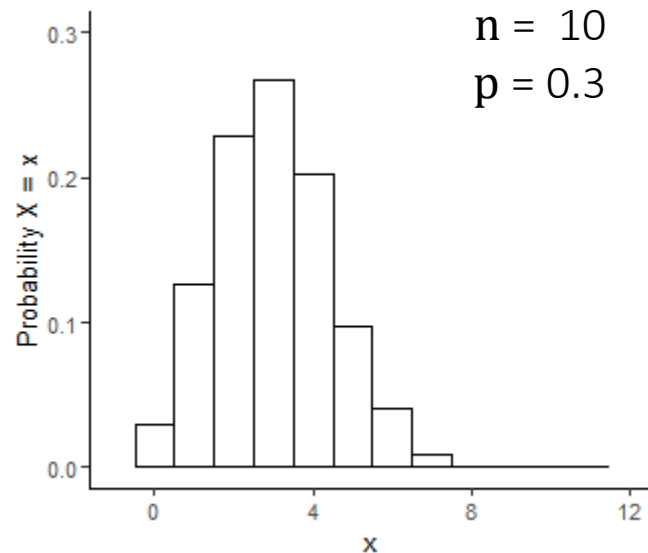
The Binomial Distribution & the Meaning of Math

- Probability mass function (PMF)

$$f(X) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- X is a discrete random variable
- Expected value $E[X] = np$

$f(X)$ is a probability $f(X) \in [0; 1]$
Variance $\text{Var}[X] = np(1 - p)$

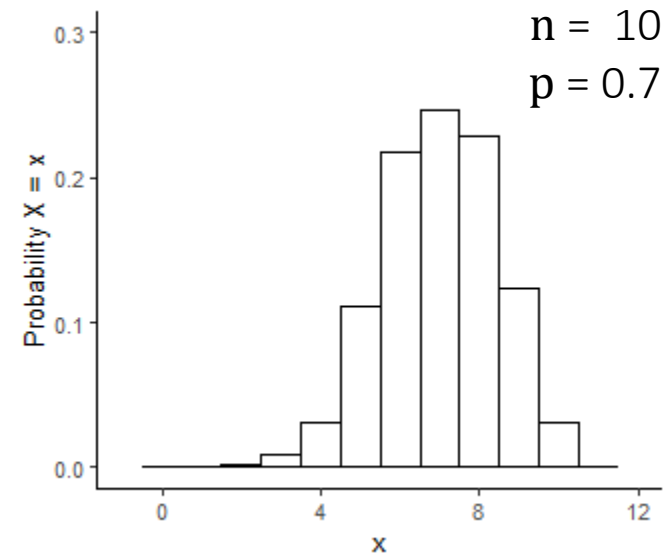
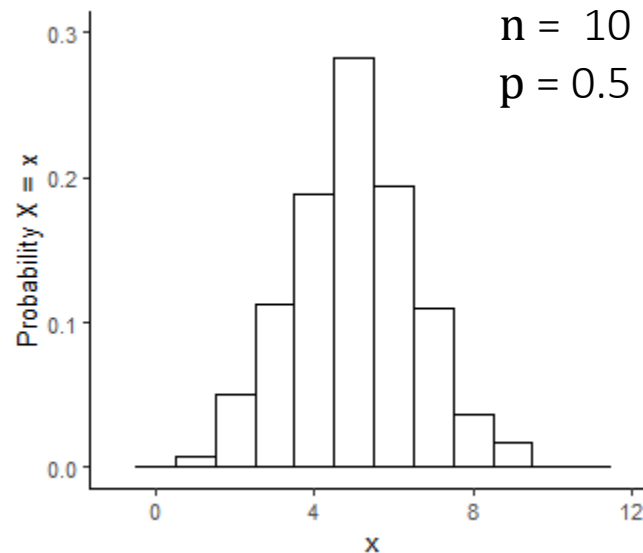
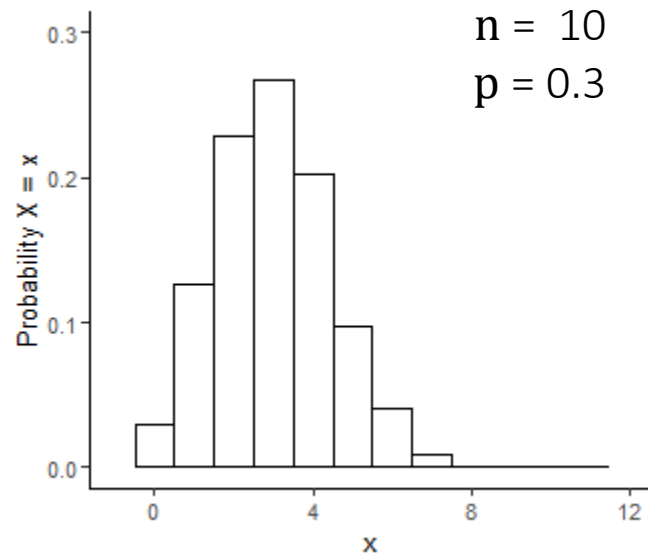


The Binomial Distribution & the Meaning of Math

- Probability mass function (PMF)

$$f(X) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- The variance is maximized for $p = 0.5$
- The binomial can be approximated with a normal distribution for very large n



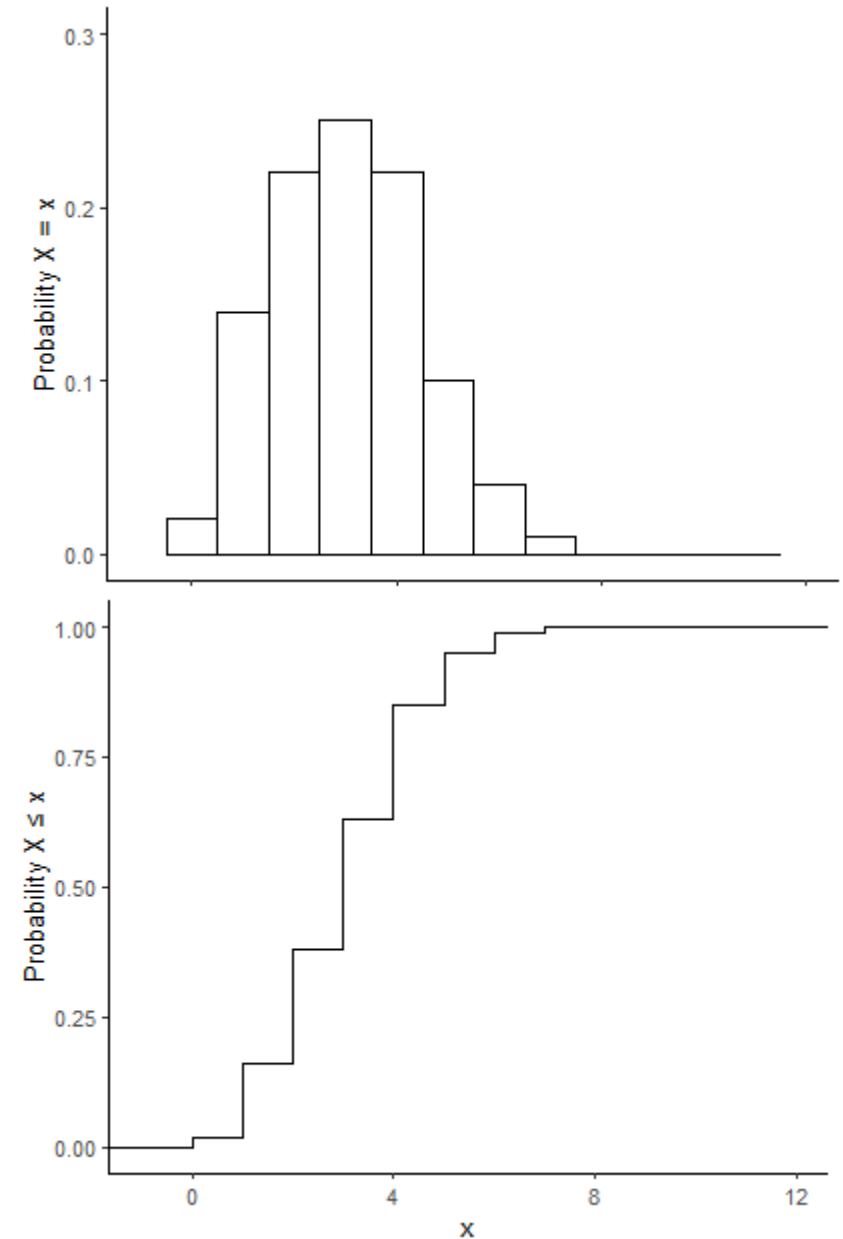
The Binomial Distribution & the Meaning of Math

- Probability mass function (PMF)

$$f(X) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- Cumulative distribution function (CMF)

$$f(X) = P(X \leq x) = \sum_{i=1}^x \binom{n}{i} p^i (1 - p)^{n-i}$$



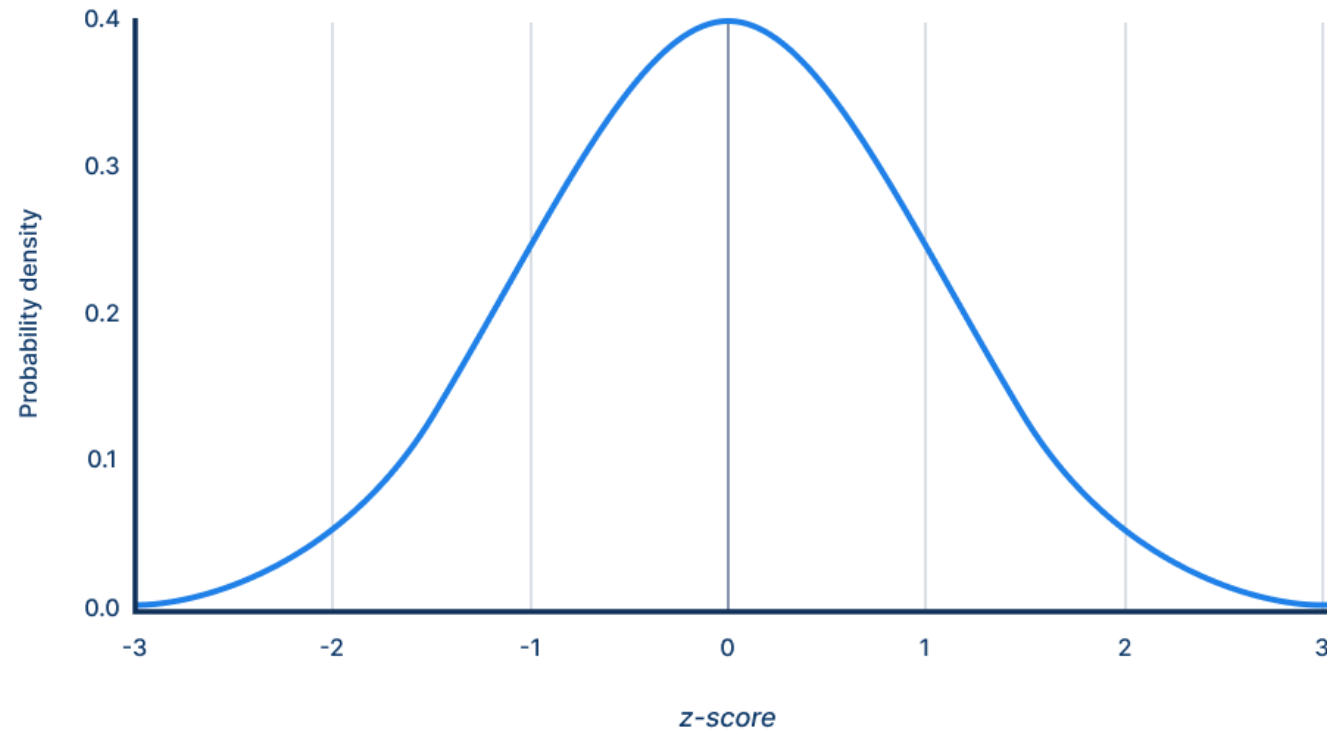
Behind the Scene: Where the Numbers Are Coming from...

The Z Distribution

Deviations from the Norm: Z Distribution and Z Score

- Z distribution = normal distribution with mean = 0 and standard deviation = 1

Standard normal distribution



The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$
- PDF and CDF of a normal distribution

$$N(\mu, \sigma) \quad P(\bar{x} = x) = \frac{1}{\sigma\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

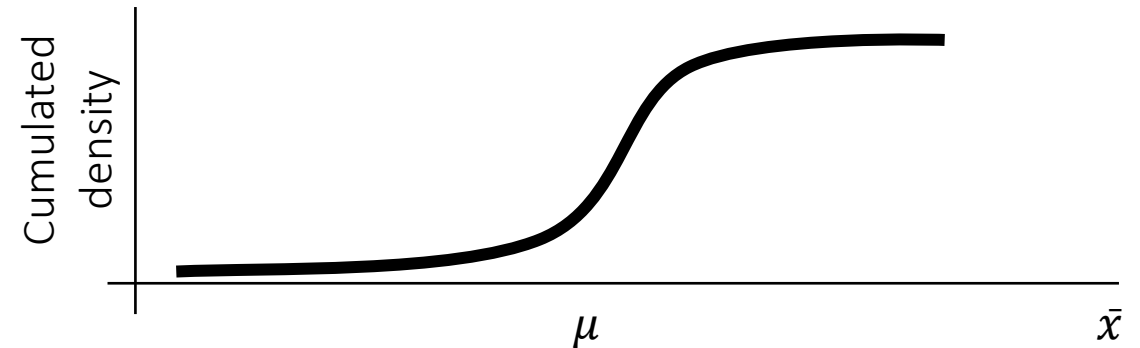
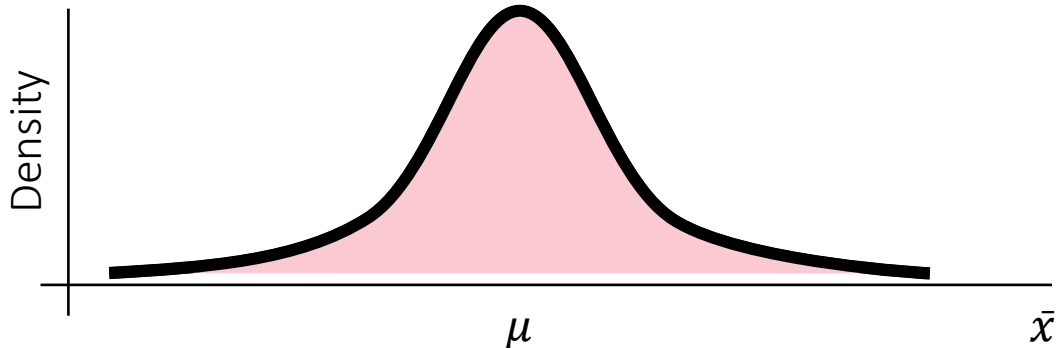
$$P(\bar{x} \leq x) = \frac{1}{\sigma\sqrt{\pi}} \int_{-\infty}^x e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Integral for all the
potential values of a
continuous variable

- PDF and CDF of the Z distribution

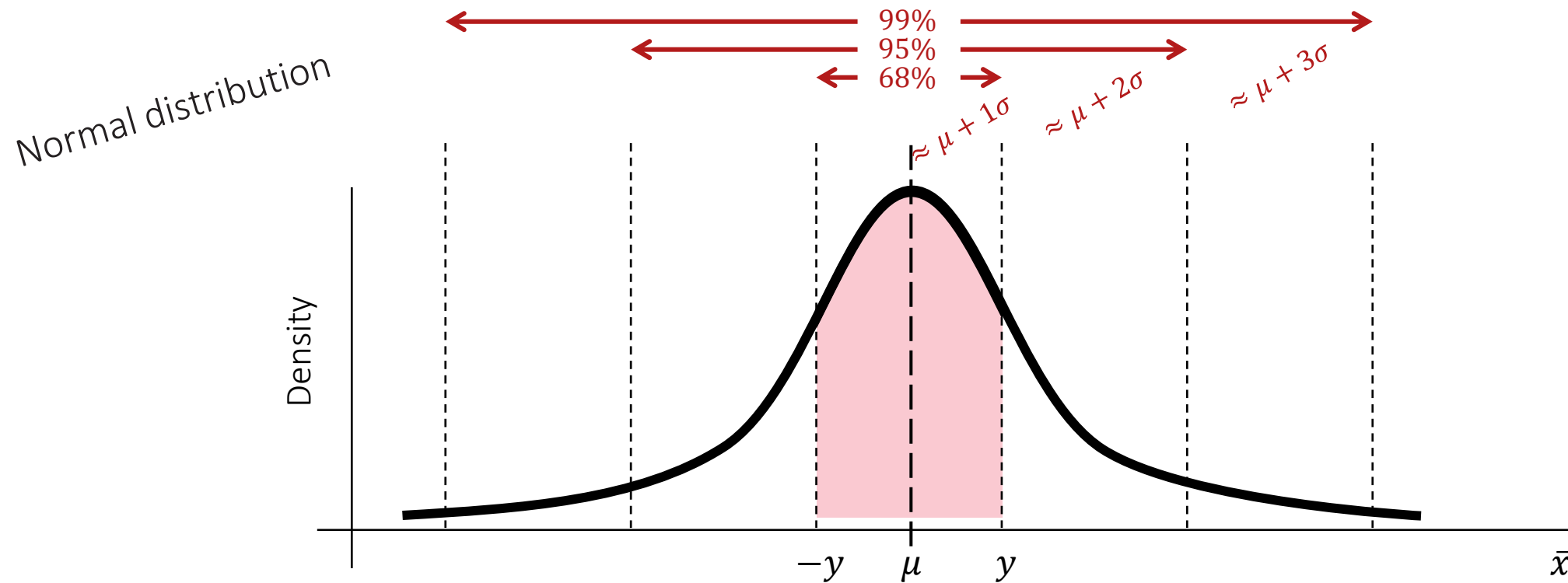
$$N(0, 1) \quad P(\bar{z} = z) = \frac{1}{\sqrt{\pi}} e^{-\frac{z^2}{2}}$$

$$P(\bar{z} \leq z) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^z e^{-\frac{z^2}{2}} dz$$



The Normal Distribution, the Z Distribution, and Intervals

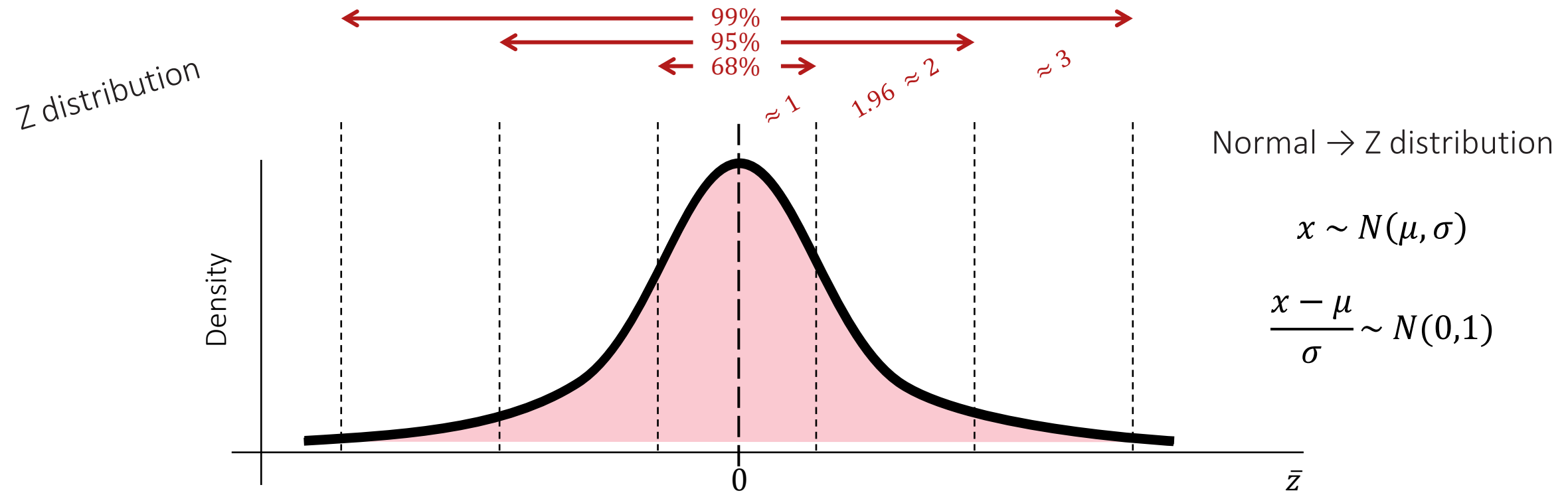
- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$



$$P(-y \leq \bar{x} \leq y) = \frac{1}{\sigma\sqrt{\pi}} \int_{-y}^y e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The Normal Distribution, the Z Distribution, and Intervals

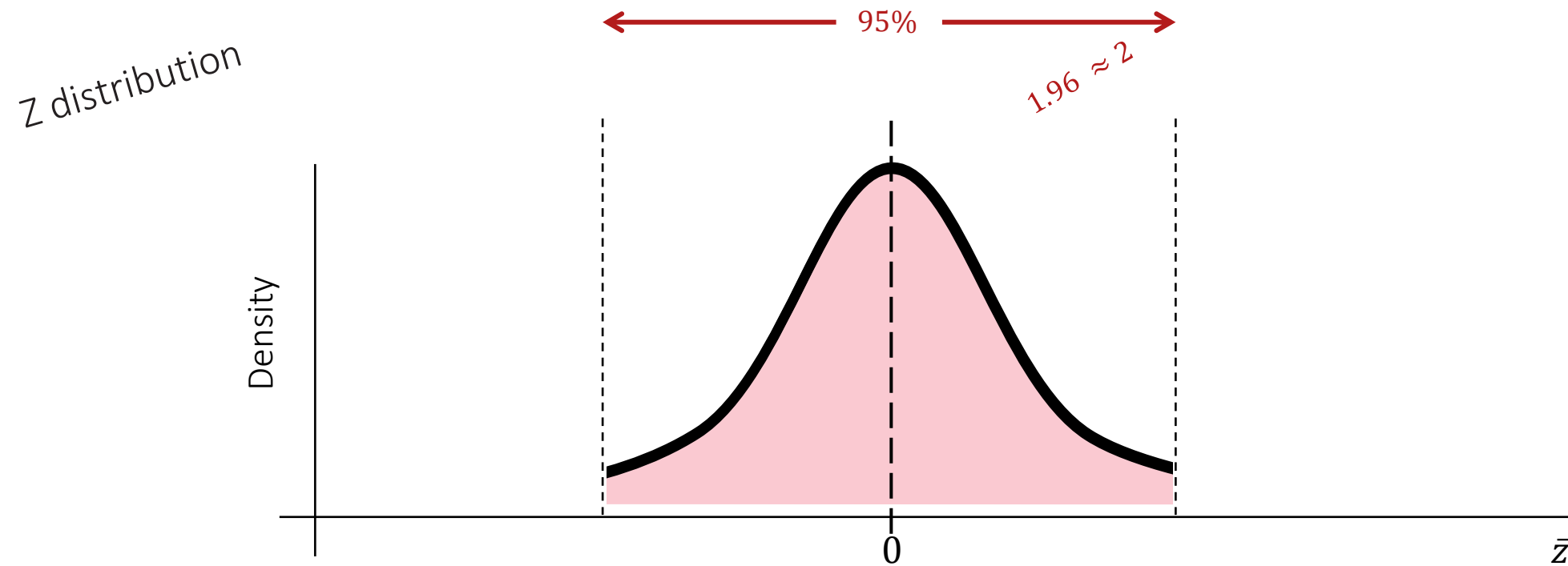
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$$P(-y \leq \bar{z} \leq y) = \frac{1}{\sqrt{\pi}} \int_{-y}^y e^{-\frac{z^2}{2}} dz$$

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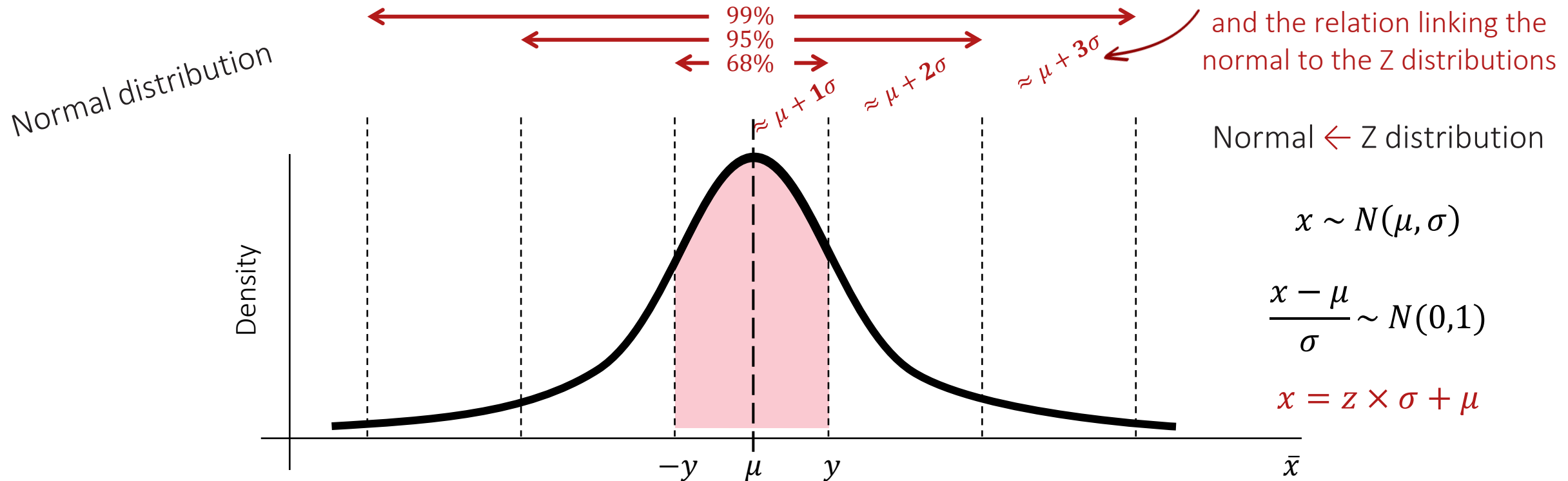


$$P(-y \leq \bar{z} \leq y) = \frac{1}{\sqrt{\pi}} \int_{-y}^y e^{-\frac{z^2}{2}} dz = 0.95 \text{ for } y = 1.96$$

This integral has no closed form, but has been tabulated

The Normal Distribution, the Z Distribution, and Intervals

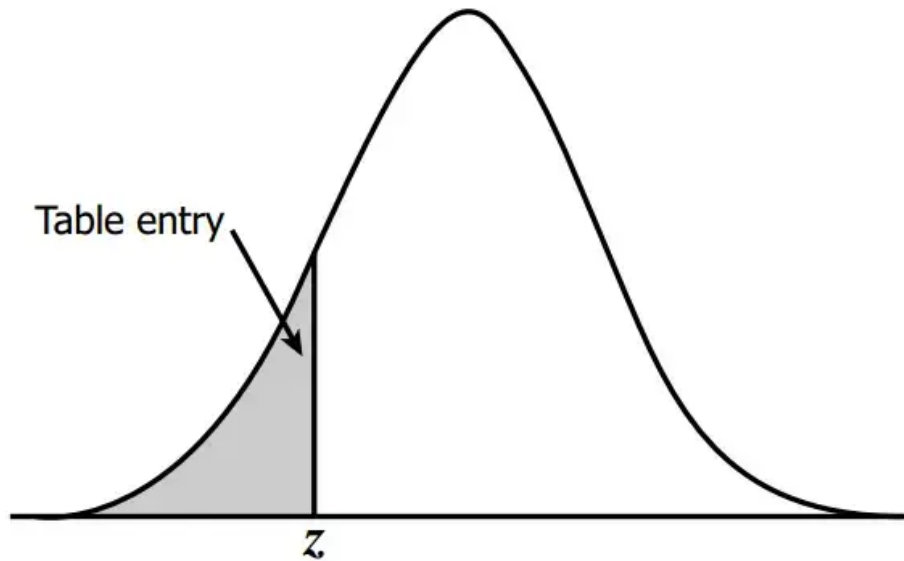
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$$P(-y \leq \bar{x} \leq y) = \frac{1}{\sigma\sqrt{\pi}} \int_{-y}^y e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The Normal Distribution, the Z Distribution, and Intervals

z-table.com



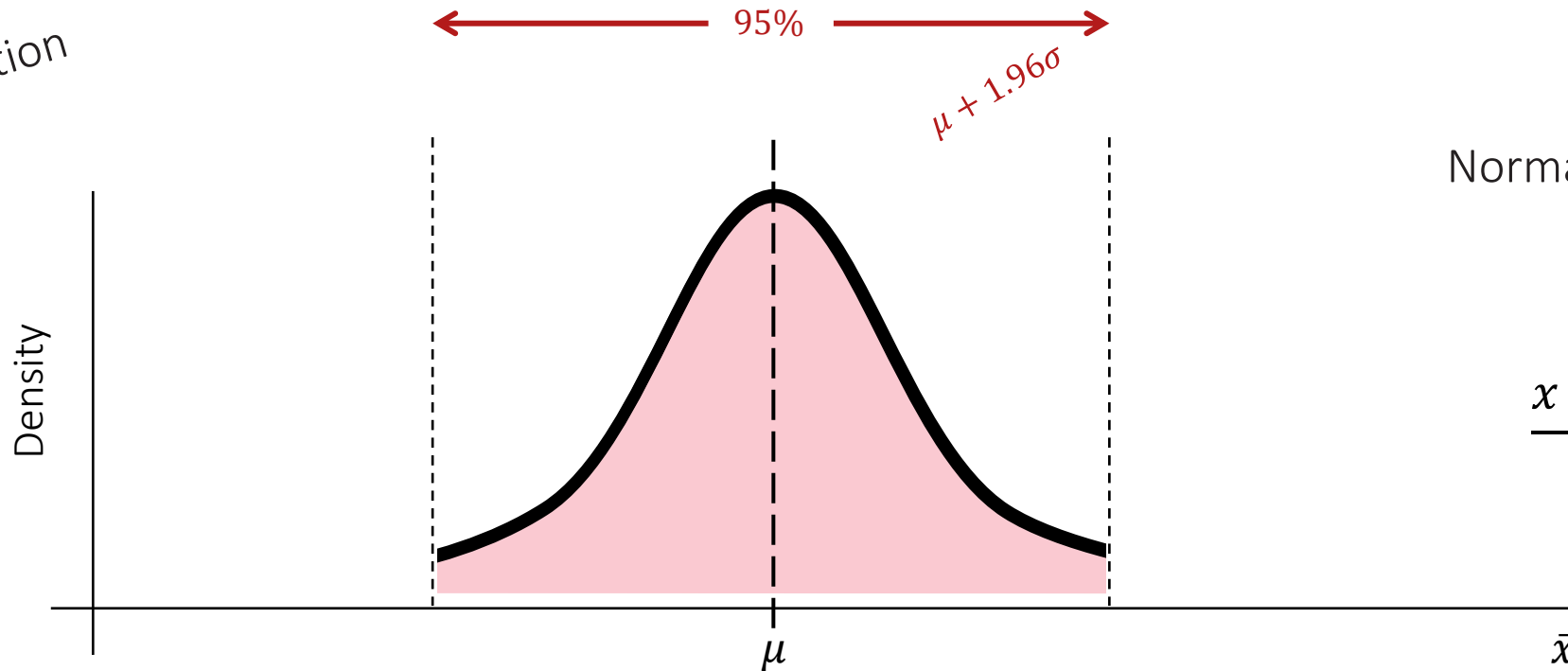
$$P(z \leq y) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^y e^{-\frac{z^2}{2}} dz$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

The Normal Distribution, the Z Distribution, and Intervals

- Challenge: we do not know the position of the population mean μ
- Good news: we know the distribution of the sample mean $\bar{x}$

Normal distribution



Normal $\rightarrow$ Z distribution

$$x \sim N(\mu, \sigma)$$

$$\frac{x - \mu}{\sigma} \sim N(0,1)$$

Confidence Intervals: Interpretation

- The same procedure applies to means and proportions

Mean

$CI = [\bar{x} - z \frac{\sigma_s}{\sqrt{n}}; \bar{x} + z \frac{\sigma_s}{\sqrt{n}}]$ with $\bar{x}$ the sample mean, σ_s the sample standard deviation, n the sample size and z the z-score for the confidence level

E.g., age at death in a population

Proportion

$CI = [\bar{p} - z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}; \bar{p} + z \sqrt{\frac{\bar{p}(1-\bar{p})}{n}}]$
with $\bar{p}$ the sample proportion, n the sample size and z the z-score for the confidence level

E.g., number of infected people in a population (prevalence)

Take Home Messages

Always report a central
tendencies with measures
of dispersion

Pick the right summary
statistics for your data (based
on their distribution)

Always report a point
estimate with measures of
uncertainty

Always accompany summary
statistics with some
visualization of the raw data

Increasing sample size
enables a higher precision
with high confidence
i.e., lower uncertainty)

We never know the true
population mean, but we know
the distribution of the sample
mean is centered around it, so
we can infer where it likely is

Before Next Time (Friday February 6, in LH3)

- Practice: [Homework #02](#) due by **Thursday 5 PM**
- Use the [Guiding Quiz #02](#) if needed
- Check you are ready for next class: [Knowledge Check #02](#) due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours
 - **Tuesdays, 09:00-10:00 AM** (Jordan, Madison and Harsh)
 - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal and Daniel)
 - If you cannot make it at these times, please contact the course assistant of your choice by email
 - in S1-122, unless indicated otherwise on Canvas